

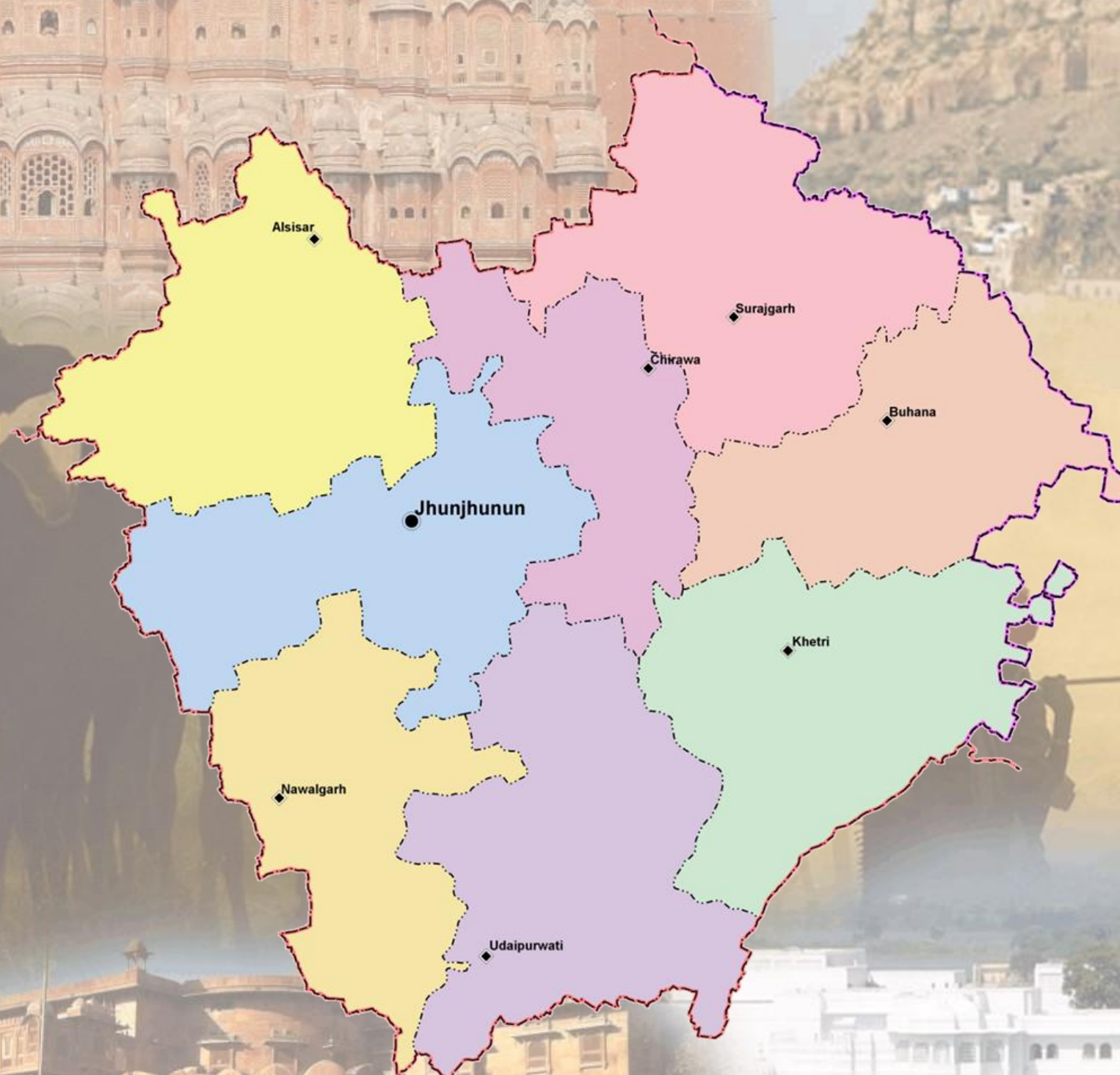


Ground Water Department,
Rajasthan

Hydrogeological Atlas of Rajasthan Jhunjhunun District



European Union
State Partnership Programme



2013



Rolta India Limited

Message



Shri Ashok Gehlot,
Hon'ble Chief Minister, Rajasthan

Ground water is an inseparable aspect in the lives of the people of Rajasthan. There are no major or perennial rivers flowing through the State. Moreover, most part of the state receives very less rainfall. Our dependence on ground water is extremely critical not only as drinking water for human beings and cattle but also as major source of irrigation.

The Government of Rajasthan is keen on sustainable utilization of the ground water. The State water policy, adopted in 2010 emphasizes on community participation in ground water management and use of information technology for widespread dissemination of data for the benefit of all sections of the society.

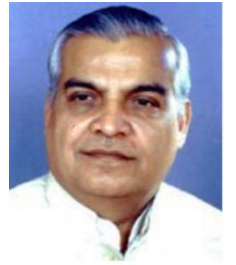
I am pleased to note that the Ground Water Department of the State has carried a detailed assessment of quality, quantity and distribution of ground water resources of the State through mapping of aquifers and using most modern tools and techniques.

It is also heartening to note that the outcome of the project is being published in the form of '*Hydrogeological Atlas of Rajasthan*'. It will incorporate the scientific findings and reliable information in various types of readily understandable GIS maps based on ground water potential viz. zone maps, depth to water level maps, ground water categorization maps, chemical quality maps, aquifer maps etc. These would be certainly found useful by experts and user agencies. The same would also help in raising the awareness levels of the various stakeholders in utilizing the precious ground water resources of the State.

I take this opportunity to congratulate the Ground Water Department officials for bringing this useful publication and wish them success in all such endeavor.

(Ashok Gehlot)

Message



Dr. Jitendra Singh,
Hon'ble Minister for Energy,
Non-Conventional Energy Resources,
PHED, GWD, Information & Public Relations

Ground water is a renewable resource but its long term sustainability depends on the balance of recharge and withdrawal ground water is added every year through rainfall and other sources. In Rajasthan, volume of water extracted for different purposes far exceeds its replenishment which has led to depletion of ground water in large parts of state.

While the surface water is visible and largely measurable, the ground water is hidden underneath. To meet the ever increasing demand of water, the state has been promoting conjunctive use of water. In order to have a state level estimation of water resources, the estimation of ground water quality and quantity is a key input. The studies being very scientific in nature it remains a challenge how to communicate the message to all stakeholders.

The publication of '*Hydrogeological Atlas of Rajasthan*' is a major step forward and will help in visualization of ground water related information in very simplified graphic and tabular forms. The effort that has gone into creation of the atlas from basic information in archives is very much appreciated.

(Jitendra Singh)

Message



Dr. Purushottam Agarwal

Principal Secretary to Government,
Public Health Engineering and Ground Water Department, Rajasthan

Ground water has been the principal source over years to meet the needs of human and cattle population of the state. Increasingly modern techniques are used for extraction of large volume of ground water which has exceeded recharge in most parts of the State. As a result, water table in most of the areas has gone down significantly. People are digging deeper and deeper to reach the ground water to meet their increasing needs. Maintaining a healthy practice of balance between recharge and withdrawal is critical for its sustainability.

The state had initiated various programs under which a systematic assessment of water resources is being carried out and all future water management plans are chalked out on the basis of sound database and historic data analysis. Increasing population, industrialization, agricultural need, power generation etc. are bound to pose challenges before us in coming years and to face them we have documented 'Water Resource Vision 2045' which highlights the short term (upto 2015) and long term (upto 2045) thrust areas and action plan which are pre-requisites for successful implementation of the State Water Policy. We are committed to achieving the objective of optimum use of every drop of scarce and precious utilizable water resource. In this regard, the Ground Water Department has also carried out a systematic assessment of ground water resources and three dimensional mapping of aquifers that gives a new insight.

I congratulate the Ground Water Department on publication of 'Hydrogeological Atlas of Rajasthan' which emphasizes on various aspects of ground water which I feel will help us in spreading the information on current scenario and consequently help in managing it better.

(Purushottam Agarwal)

Foreword



Arno Schaefer

Minister Counselor, Head of Operations

The Government of Rajasthan has entered into an agreement with the European Union through the State Partnership Programme for implementing state-wide water sector reform leading to sustainable integrated water resources management in Rajasthan, and supporting the Panchayat Raj institutions in executing their responsibilities in equitable access to safe, adequate, affordable and sustainable water supply as well as conservation, stabilization and replenishment of surface and ground water.

Ground water is the primary source for drinking, domestic, agriculture and industries and other purposes in Rajasthan. The stage of ground water development is close to 135% i.e., if the net annual ground availability in Rajasthan is about 10.7 billion cubic meter, the annual draft is 14.5 billion cubic meter. As a result of this, the water levels have been depleting alarmingly. The ground water in 166 out of 249 blocks of the state fall under the category of over-exploited. This indiscriminate and sometime exploitative use of ground water has led to questions regarding its sustainability.

This publication '*Hydrogeological Atlas of Rajasthan*' is the culmination of a long term effort of collation, integration, interpretation and analysis of vast archive of historic data obtained by Rajasthan Ground Water Department. The atlas is in three parts, (1) Statewide Atlas, which shows the state of ground water, water quality, current stage of exploitation, aquifer mapping and basic information on administrative setup, climate, geology and geomorphology; (2) A detailed series, wherein compilation of information and district level thematic maps and all 33 districts have been separately presented and (3) Compilation of information at the River Basin level. In addition to the thematic maps, a large number of cross sections and the 3 dimensional features of aquifers have also been presented.

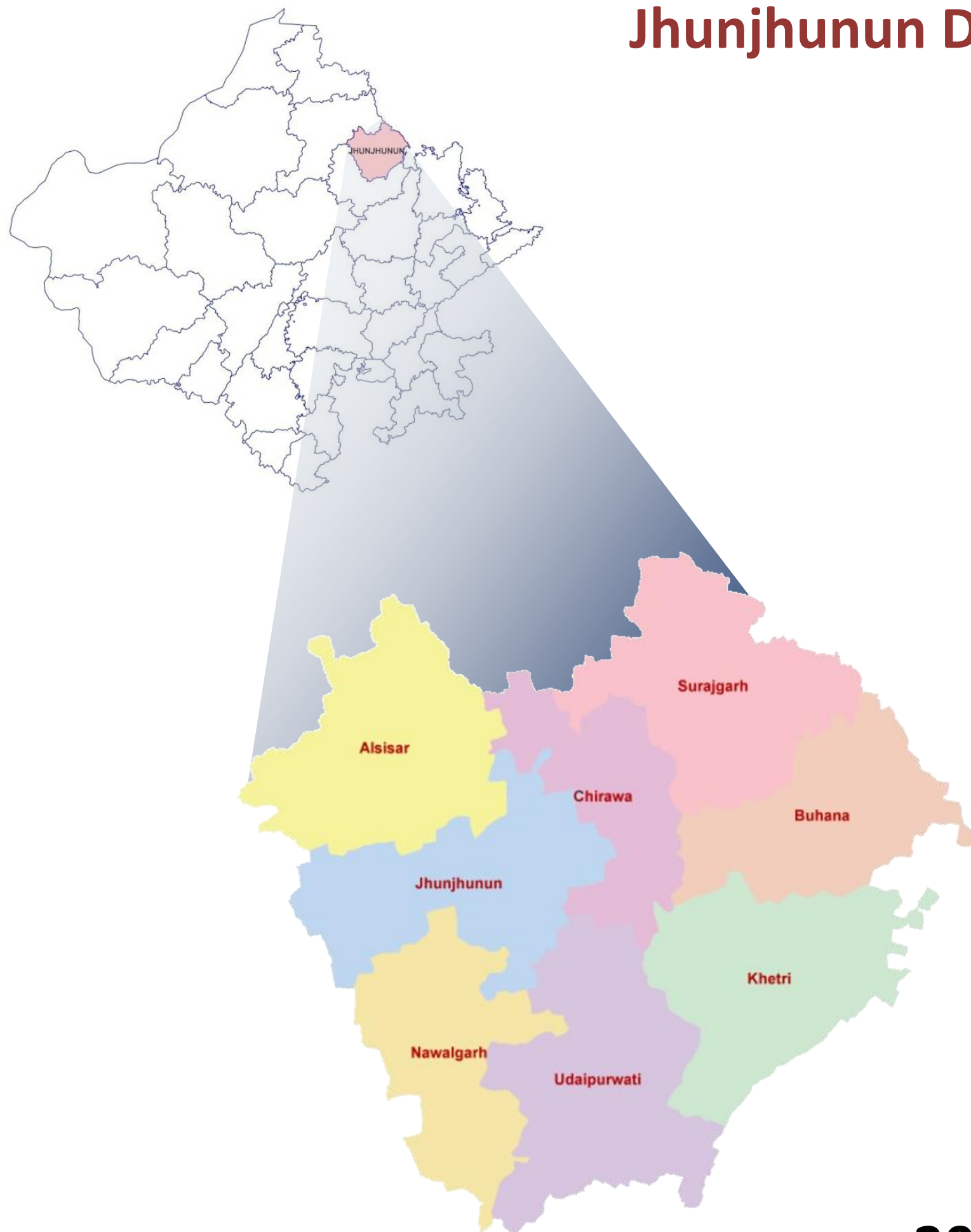
I am convinced that the Atlas will contribute to better understand ground water and help managerial and future planning. The current Geographic Information System (GIS) based study is a pioneering step in India and will pave the way for similar approach to be adopted by other Indian States.

I appreciate very much the efforts made by the Ground Water Department, Government of Rajasthan, The European Union Technical Assistance Team and M/s Rolta India Limited for undertaking this study and compiling this Atlas under the India-EU State partnership Programme.

(Arno Schaefer)

Hydrogeological Atlas of Rajasthan

Jhunjhunun District



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ADMINISTRATIVE SETUP

DISTRICT – JHUNJHUNUN

Location:

Jhunjhunun district is located in the northern part of Rajasthan. It is bounded in the northwest by Churu district, in the east state of Haryana and in the southwest by Sikar. It stretches between 27° 38' 13.88" to 28° 31' 11.09" North latitude and 75° 01' 30.74" to 76° 06' 01.47" East longitude covering an approximate area of 5,911.1 sq kms. Major part of the district does not have a systematic drainage system because, except for a strip in the centre of the district running northwest to southeast, which is part of Shekhawati River Basin, the remaining part of the district both in the east and west the region forms part of a so called "Outside Basin".

Administrative Set-up:

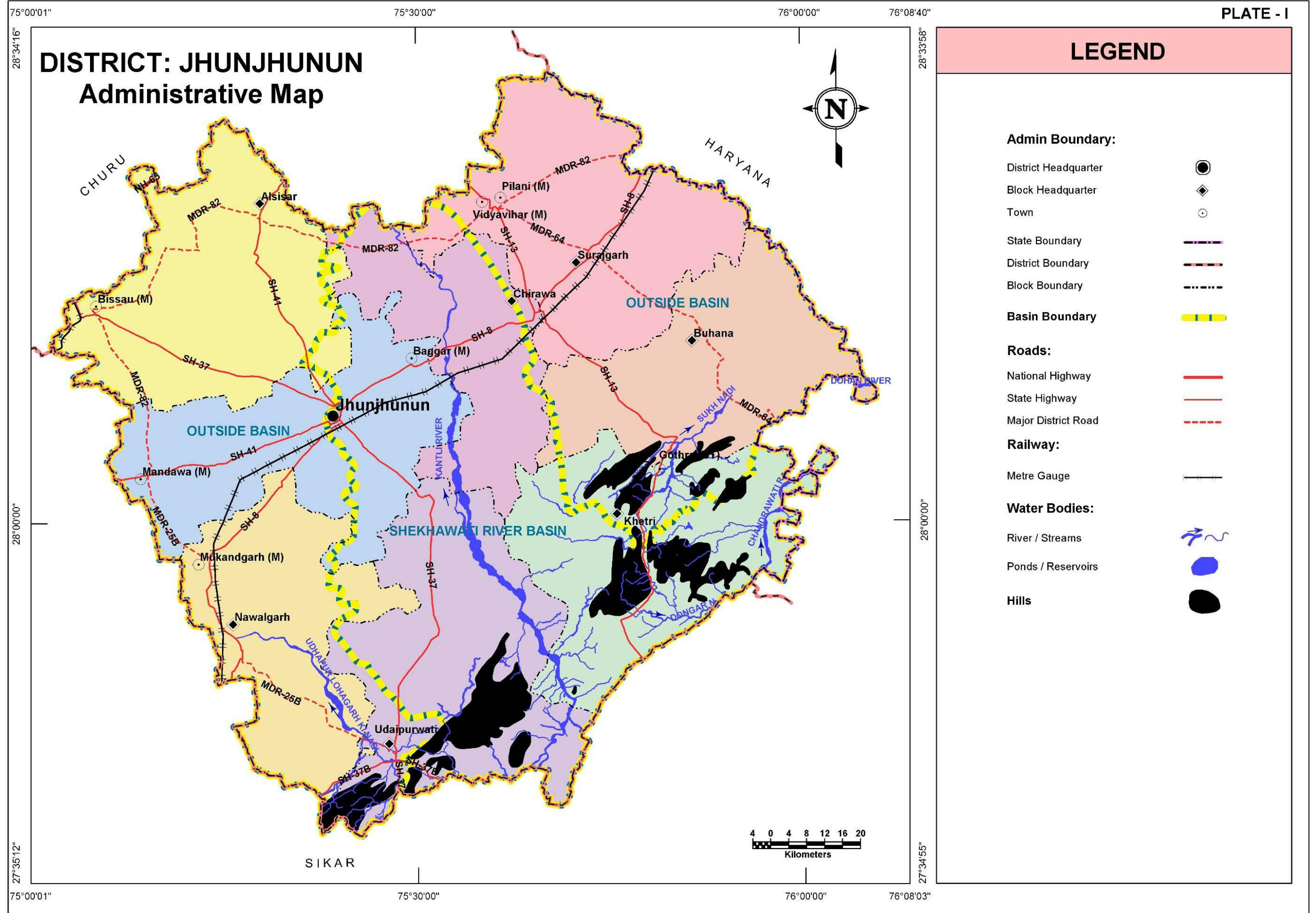
Jhunjhunun district is administratively divided into eight blocks. The following table summarizes the basic statistics of the district at block level.

S. No.	Block Name	Population (Based on 2001 census)	Area (sq km)	% of District Area	Total Number of Towns and Villages
1	Alsisar	1,62,549	818.9	13.8	126
2	Buhana	2,04,411	652.4	11.0	126
3	Chirawa	1,90,070	522.0	8.8	85
4	Jhunjhunun	3,12,666	764.3	12.9	148
5	Khetri	2,53,676	813.5	13.7	88
6	Nawalgarh	2,91,919	643.0	10.8	92
7	Surajgarh	2,35,275	815.5	14.0	125
8	Udaipurwati	2,63,123	881.5	15.0	82
Total		19,13,689	5,911.1	100.0	872

Jhunjhunun district has 872 towns and villages, of which eight are block headquarters as well.

Climate:

The climate of the district can be classified as largely, arid. It is characterized by very hot summers and very cold winters with generally poor but occasionally good rainfall during southwest monsoon period. In May and June, the maximum temperature may sometimes go up to 48 °C. Winter temperatures drop down to 3°C to 4°C. The potential evapotranspiration rates are quite high, especially during May and June. Average annual rainfall of this district is about 440.0 mm.



TOPOGRAPHY

The topography of the district displays hilly area in the eastern and southeastern part which belong Aravalli range, running in NE-SW direction. Rest of the district has broad undulating plains. The general slope of the terrain in the district is from southeast to northwest. The area is drained mainly by Shekhawati, Kantli and by the Chandrawati Rivers with their tributaries like Udhapur Ohagarh, Dongar, Sukh etc. The general topographic elevation in the district is between 250 m to 500 m above mean sea level in most of the blocks. Elevation ranges from a minimum of 259.6 m above mean sea level in Surajgarh block in the NE part of the district to maximum of 1,035.0 m above mean sea level In Nawalgarh in SW part of the district.

Table: Block wise minimum and maximum elevation

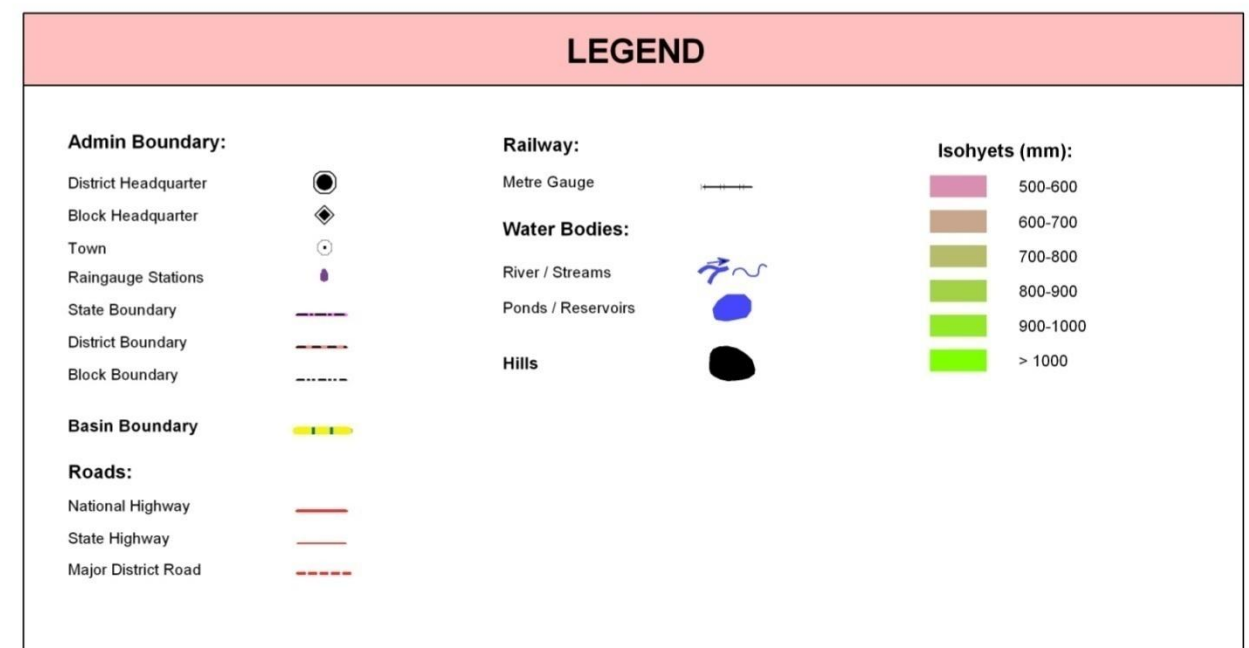
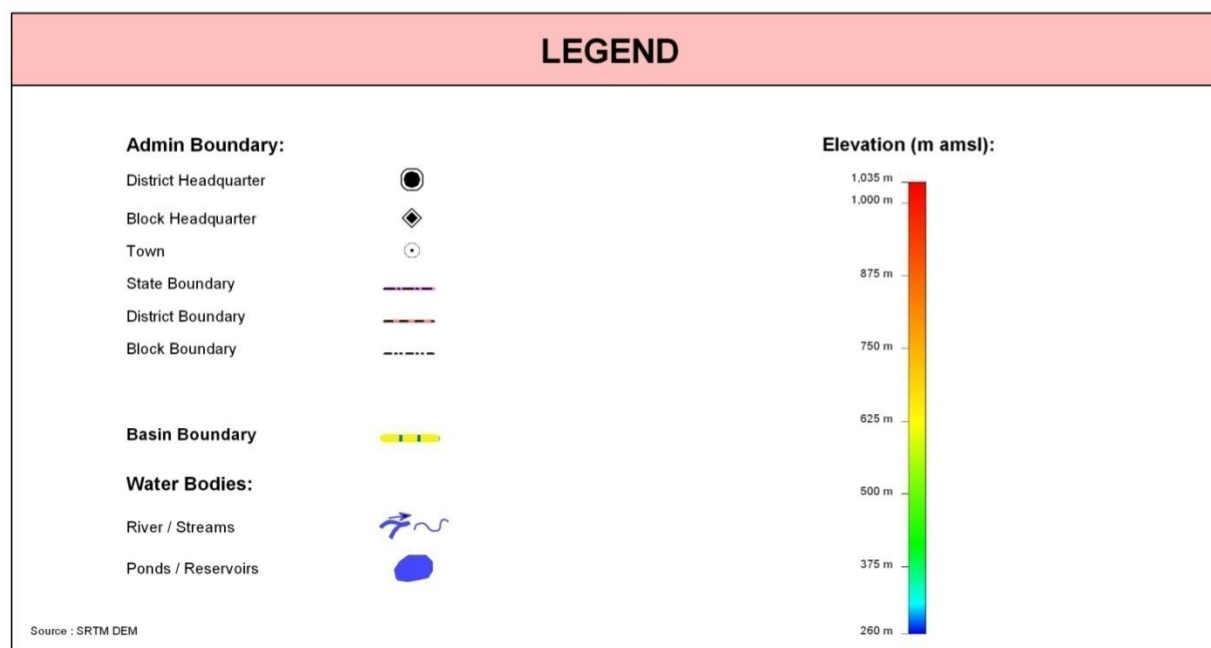
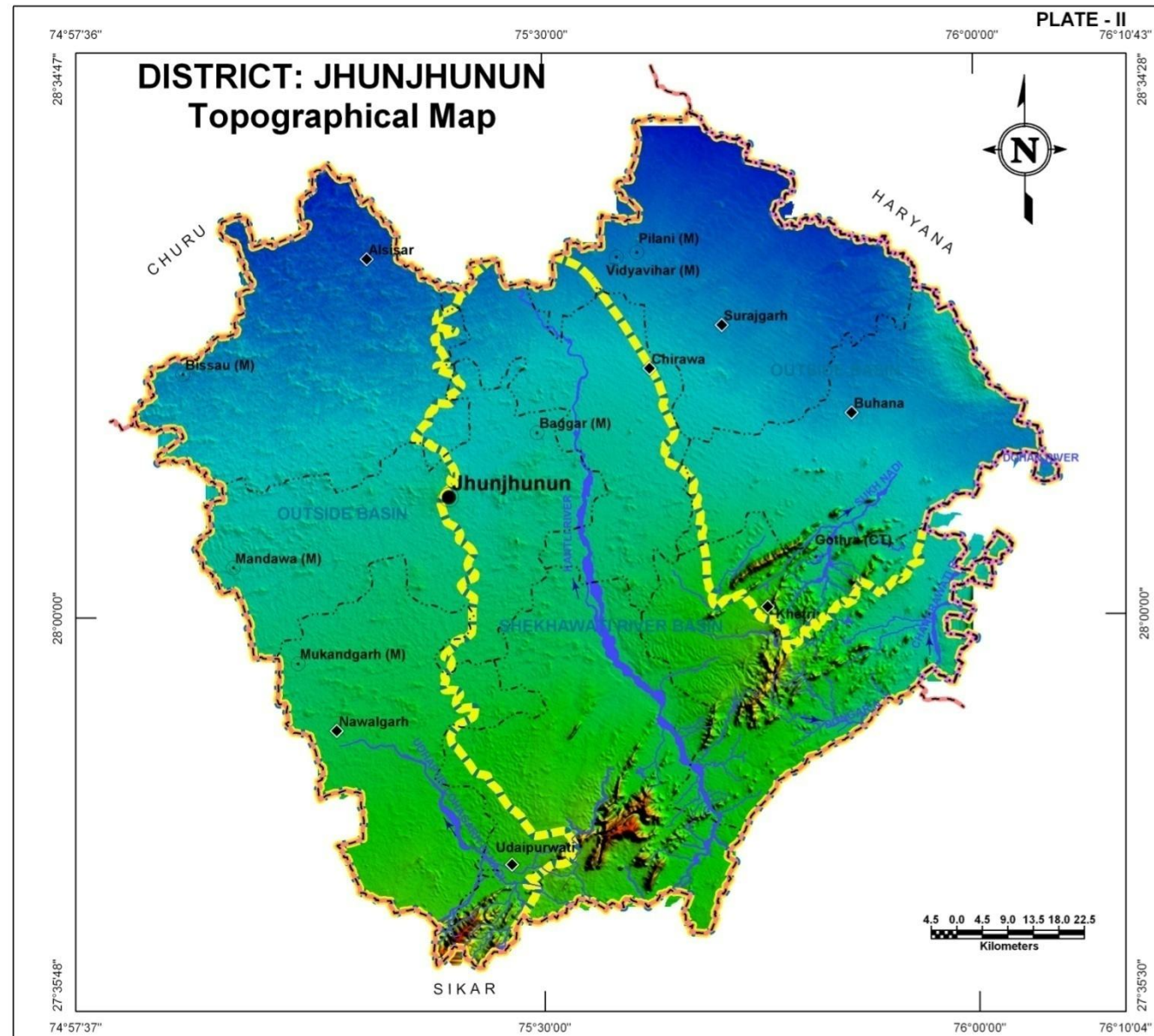
S. No.	Block Name	Min. Elevation (m amsl)	Max. Elevation (m amsl)
1	Alsisar	259.7	339.0
2	Buhana	280.8	644.2
3	Chirawa	282.7	411.0
4	Jhunjhunun	300.9	474.3
5	Khetri	307.1	774.2
6	Nawalgarh	338.7	1,035.0
7	Surajgarh	259.6	326.7
8	Udaipurwati	351.3	1,017.6

RAINFALL

Rainfall received in the year 2010 has been very good as compared to other years, because as mentioned above, the district generally receives low rainfall. The general distribution of rainfall across can be visualized from isohyets presented in the Plate – III. Rainfall is highest in the areas closer to hills in the southeastern part of the district as well as in the northwestern part where high rainfall pockets of about 1000mm are seen. The eastern and southwestern parts of the district show low rainfalls and interestingly these areas correspond to Outside basin. The annual average rainfall is 772.3 mm based on the data of available blocks.

Table: Block wise annual rainfall statistics (derived from year 2010 meteorological station data)

Block Name	Minimum Annual Rainfall (mm)	Maximum Annual Rainfall (mm)	Average Annual Rainfall (mm)
Alsisar	697.2	987.2	854.7
Buhana	569.4	816.9	650.5
Chirawa	567.2	894.4	734.0
Jhunjhunun	699.1	848.8	778.9
Khetri	680.9	1,125.5	913.7
Nawalgarh	657.5	838.1	740.7
Surajgarh	568.8	785.0	617.5
Udaipurwati	746.2	1,129.4	888.0



GEOLOGY

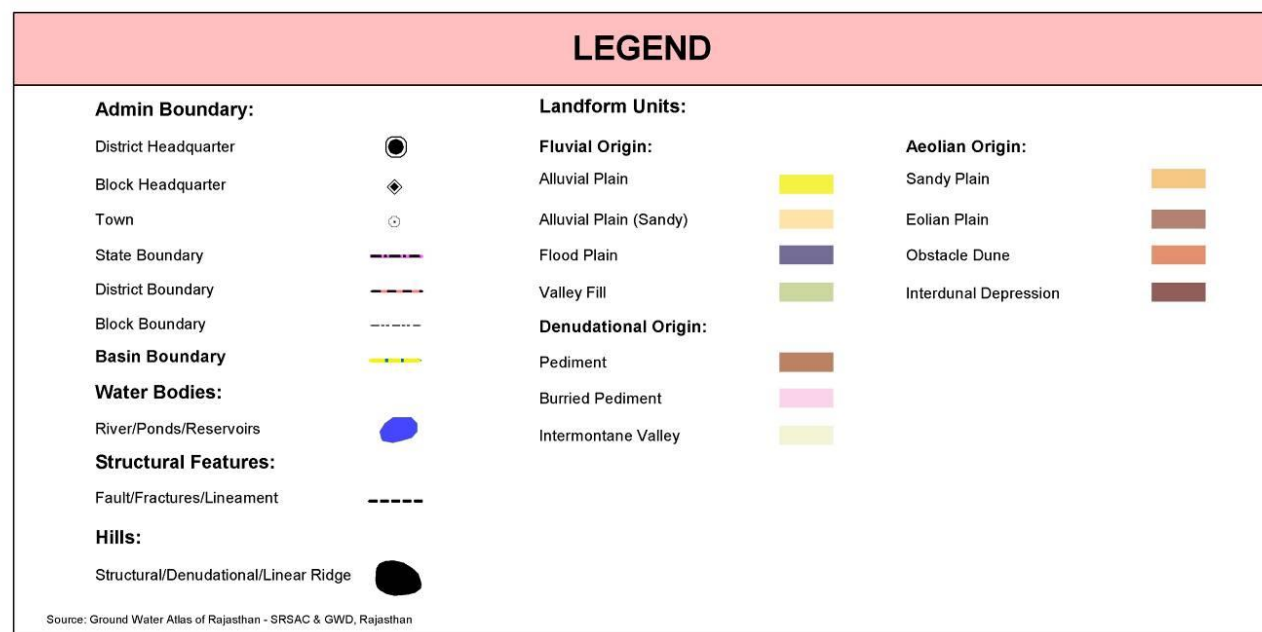
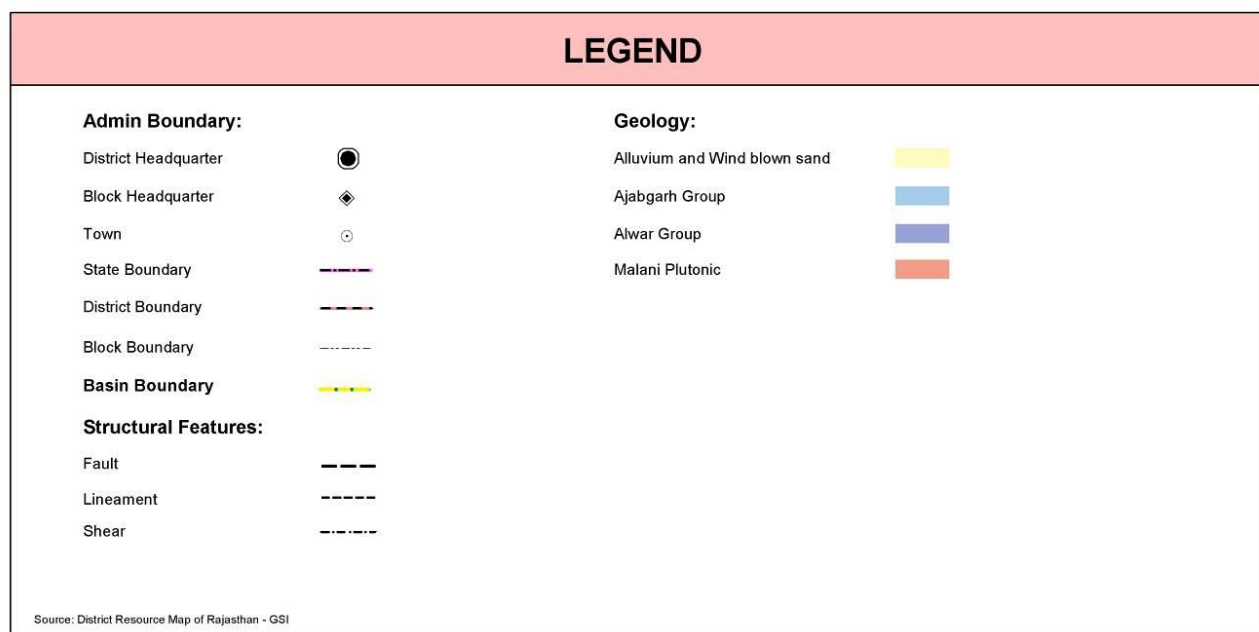
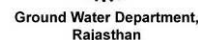
Geologically, the district is mostly covered by blown sand. Apart from these, small isolated outcrop of the rocks of Delhi Super Group and Malani igneous suite are found in Khetri, Udaipurwati, Buhana and Jhunjhun blocks of the district. The Alwar Groups of rocks are exposed in the southeastern part (around Udaipurwati) and northeastern part (around Khetri), with Ajabgarh Group of rocks, but here they as thin and narrow outcrops only. The Alwar Groups are represented by quartzite, schist, grit, arkose etc. These have been intruded by Post Delhi intrusives such as amphibolites, granite, pegmatite, quartz veins etc. The Ajabgarhs are represented by phyllites, biotite schists, calc gneisses etc. intruded by Post Delhi intrusives viz. amphibolite, granite, albitites, pegmatites, epidiorite, quartz veins etc. The basic intrusives include epidiorite, diorite amphibolite etc.

Super Group	Group	Formation
	Recent to Sub-Recent	Alluvium, Aeolian sand
	Post –Delhi	Pegmatite- Erinpura Granite
	Intrusives	Epidiorite, dolerite, albitite, granite (Malani Igneous Suite) amphibolite.
Delhi	Ajabgarh	Phyllites, phyllites interbedded with quartzites, biotiteschist, calcgneissesetc
	Alwar	Quartzite, schist, flaggy quartzite, grit, arkose, felspathic quartzite etc.
	Malani Plutonic	

GEOMORPHOLOGY

Table: Geomorphologic units, their description and distribution

Origin	Landform Unit	Description
Aeolian	Eolian Plain	Formed by aeolian activity, with sand dunes of varying height, size, and slope. Long stretches of sand sheet. Gently sloping flat to undulating plain, comprised of fine to medium grained sand and silt. Also scattered xerophytic vegetation.
	Interdunal Depression	Slightly depressed area in between the dunal complex showing moisture and fine sediments.
	Obstacle Dune	Formed on windward/leeward sides of obstacle like isolated hills or continuous chain of hills, dune to obstruction in path of sand laden winds. Badly dissected well cemented and vegetated.
	Sandy Plain	Formed of aeolian activity, wind-blown sand with gentle sloping to undulating plain, comprising of coarse sand, fine sand, silt and clay.
Denudational	Buried Pediment	Pediment covers essentially with relatively thicker alluvial, colluvial or weathered materials.
	Intermontane Valley	Depression between mountains, generally broad & linear, filled with colluvial deposits.
	Pediment	Broad gently sloping rock flooring, erosional surface of low relief between hill and plain, comprised of varied lithology, criss-crossed by fractures and faults.
Fluvial	Alluvial Plain	Mainly undulating landscape formed due to fluvial activity, comprising of gravels, sand, silt and clay. Terrain mainly undulating, produced by extensive deposition of alluvium.
	Alluvial Plain (Sandy)	Flat to gentle undulating plain formed due to fluvial activity, mainly consists of gravels, sand, silt and clay with unconsolidated material of varying lithology, predominantly sand along river.
	Flood Plain	The surface or strip of relatively smooth land adjacent to a river channel formed by river and covered with water when river over flows its bank. Normally subject to periodic flooding.
	Valley Fill	Formed by fluvial activity, usually at lower topographic locations, comprising of boulders, cobbles, pebbles, gravels, sand, silt and clay. The unit has consolidated sediment deposits.
Hills	Denudational, Structural Hill, Linear Ridge	Steep sided, relict hills undergone denudation, comprising of varying lithology with joints, fractures and lineaments. Linear to arcuate hills showing definite trend-lines with varying lithology associated with folding, faulting etc. Long narrow low-lying ridge usually barren, having high run off may form over varying lithology with controlled strike.



AQUIFERS

DISTRICT – JHUNJHUNUN

Aquifers in the district are formed in alluvium and quartzites. Alluvium constitutes principal aquifer material with Older Alluvium contributing to 76% of district's aquifer area and Younger Alluvium forming another 6% of aquifers. Fractured and weathered quartzite adjoining hills in the southeastern part of the district forms about 14% of aquifers. The quality of water in the western part of the alluvial aquifers is not as good as that of the eastern part.

Table: aquifer potential zones their area and their description

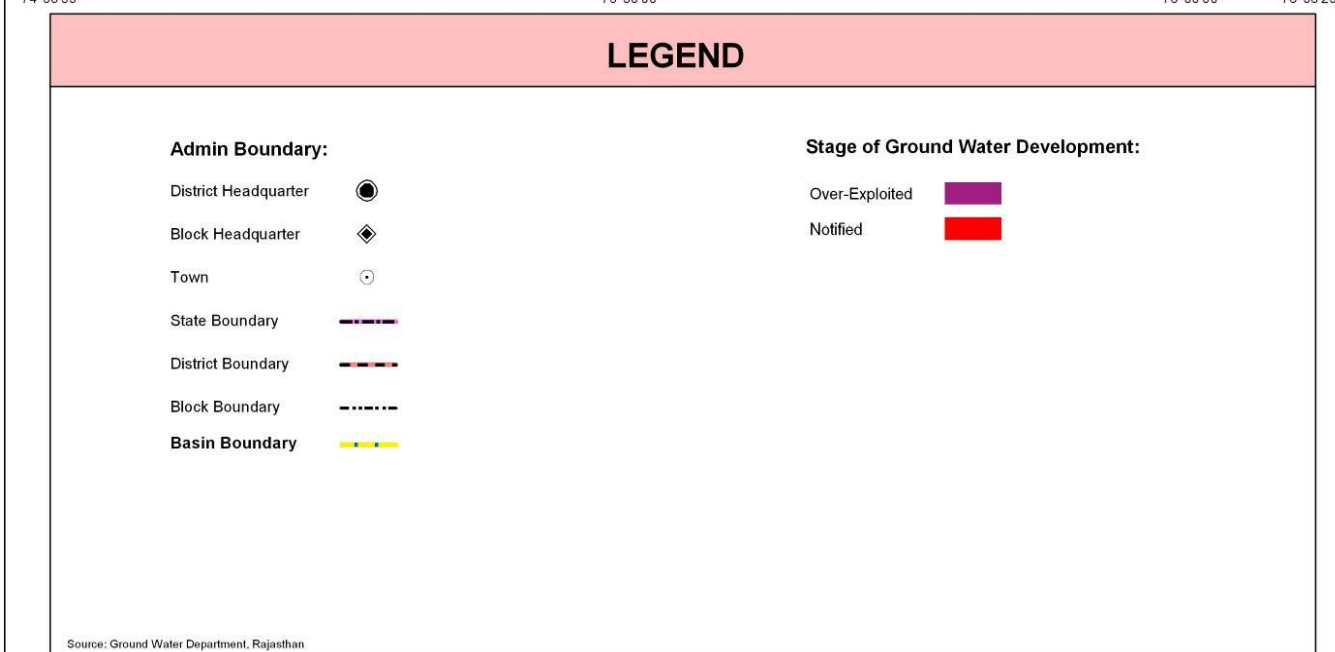
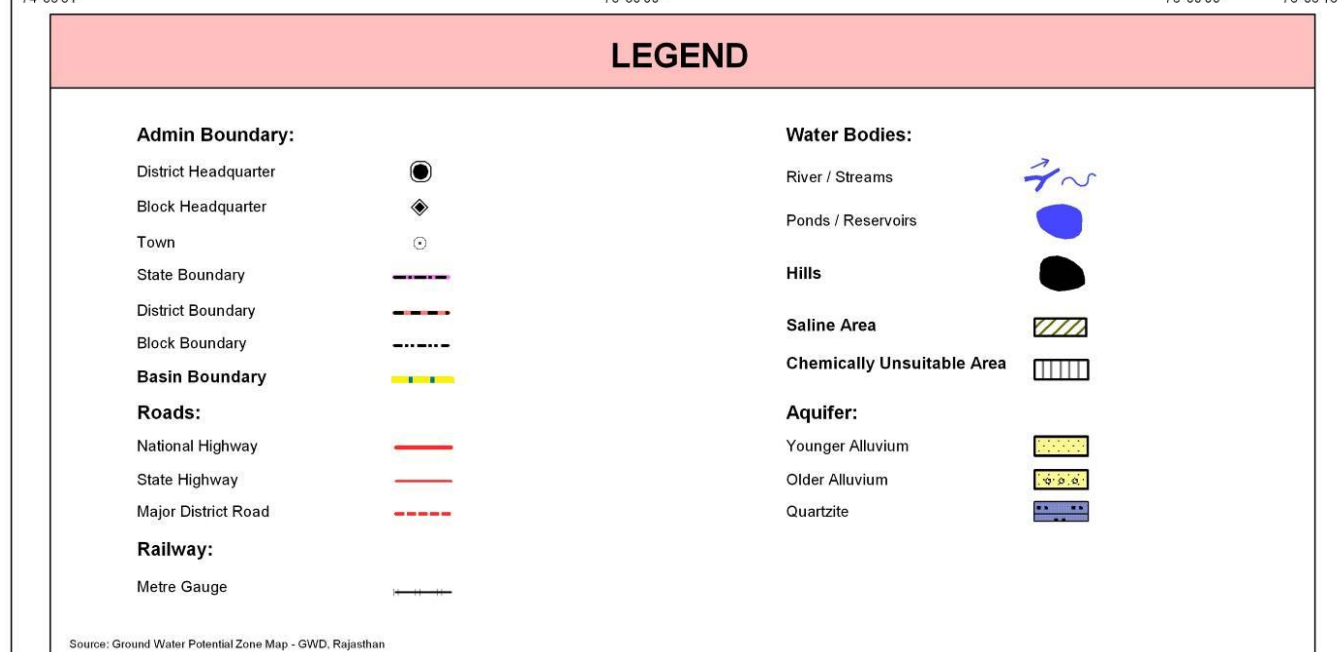
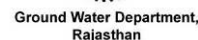
Aquifer in Potential Zone	Area (sq km)	% age of district	Description of the unit/Occurrence
Younger Alluvium	355.8	6.0	It is largely constituted of Aeolian and Fluvial sand, silt, clay, gravel and pebbles in varying proportions.
Older Alluvium	4,474.8	75.7	This litho unit comprises of mixture of heterogeneous fine to medium grained sand, silt and kankar.
Quartzite	802.7	13.6	Medium to coarse grained and varies from feldspathic grit to sericitic quartzite.
Hills	277.8	4.7	
Total	5,911.1	100.0	

STAGE OF GROUND WATER DEVELOPMENT

The ground water resource assessment studies reveal that two of the blocks in the district fall within 'Over Exploited' category. This implies that the status of development is already around 100%. The remaining six districts are even more stressed as they are in 'Notified' category where no further development of ground water is permitted.

Categorization on the basis of stage of development of ground water	Block Name
Over Exploited	Alsisar, Khetri
Notified	Nawalgarh, Udaipurwati, Jhunjhunun, Chirawa, Surajgarh, Buhana

Basis for categorization: Ground water development >100% - Over-Exploited. In Notified blocks development of GW is not permitted any more.



LOCATION OF EXPLORATORY AND GROUND WATER MONITORING WELLS

DISTRICT – JHUNJHUNUN

There is a well distributed network of exploratory wells (170) and ground water monitoring stations (135) in the district owned by RGWD (85 and 91 respectively) and CGWB (85 and 44 respectively) in the district. The exploratory wells have formed the basis for delineation of subsurface aquifer distribution scenario in three dimensions. Benchmarking and optimization studies suggest that existing ground water level network does not require any additional strengthening but ground water quality monitoring networks need to be strengthened further by adding 153 additional wells to existing network.

Table: Block wise count of wells (existing and recommended)

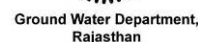
Block Name	Exploratory Wells			Ground Water Monitoring Stations			Recommended additional wells for optimization of monitoring network	
	CGWB	RGWD	Total	CGWB	RGWD	Total	Water Level	Water Quality
Alsisar	11	11	22	7	12	19	0	10
Buhana	17	11	28	10	10	20	0	20
Chirawa	10	10	20	4	8	12	0	15
Jhunjhunun	10	18	28	5	13	18	0	29
Khetri	3	9	12	6	10	16	0	8
Nawalgarh	17	9	26	4	8	12	0	25
Surajgarh	8	9	17	3	12	15	0	27
Udaipurwati	9	8	17	5	18	23	0	19
Total	85	85	170	44	91	135	0	153

DEPTH TO WATER LEVEL (PRE MONSOON – 2010)

Depth to water level varies significantly from less than 20m below ground level to more than 70mbgl. North and northeastern part of the district i.e., Chirawa-Surajgarh-Buhana region shows deeper water levels of 40 - 60m bgl in general, and reaching upto >70m bgl. Western side of the district, the water level is moderately deep (around 20-40m bgl) but in down south (Udaipurwati block), the water level is quite shallow occurring at less than 20m bgl.

Depth to water level (m bgl)	Block wise area coverage (sq km) *								Total Area (sq km)
	Alsisar	Buhana	Chirawa	Jhunjhunun	Khetri	Nawalgarh	Surajgarh	Udaipurwati	
< 20	-	-	-	-	0.1	-	-	0.6	0.7
20-30	33.3	4.5	-	-	210.2	7.0	-	78.0	333.0
30-40	454.1	77.4	34.1	7.2	327.3	82.4	-	486.0	1,468.5
40-50	283.3	129.0	101.0	339.7	78.9	134.8	-	194.2	1,260.9
50-60	48.2	138.2	187.6	395.6	32.5	411.2	327.6	0.3	1,541.2
60-70	-	262.7	199.2	21.8	27.4	0.4	487.9	-	999.4
> 70	-	29.6	-	-	-	-	-	-	29.6
Total	818.9	641.4	521.9	764.3	676.4	635.8	815.5	759.1	5,633.3

* The area covered in the derived maps is less than the total district area since the hills have been excluded from interpolation/contouring.



WATER TABLE ELEVATION (PRE MONSOON – 2010)

DISTRICT – JHUNJHUNUN

General Water table elevation of the district is from South to North. The South and Southeastern part of the district falls under the higher elevation ranges whereas northeast part comes under the lower elevation ranges. Water table elevation highest reaching up to >440m amsl in South part (Udaipurwati Block) of the district whereas minimum elevation (<180m amsl) in the Chirawa and Surajgarh blocks.

Table: Block wise area covered in each water table elevation range

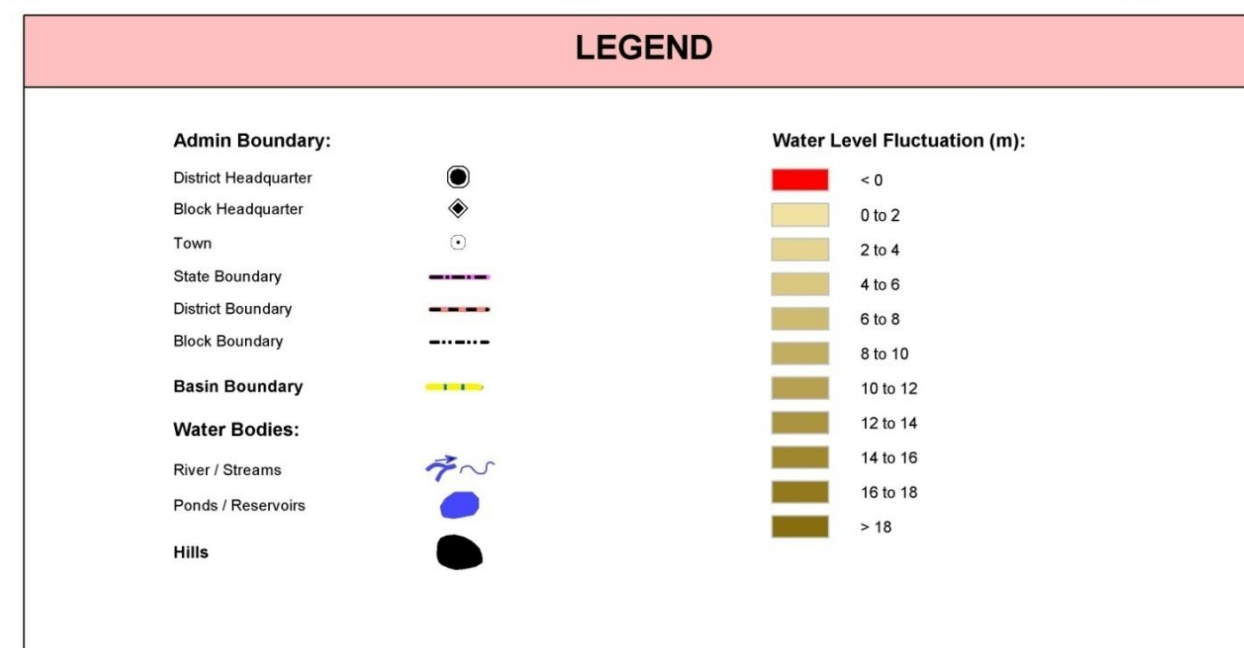
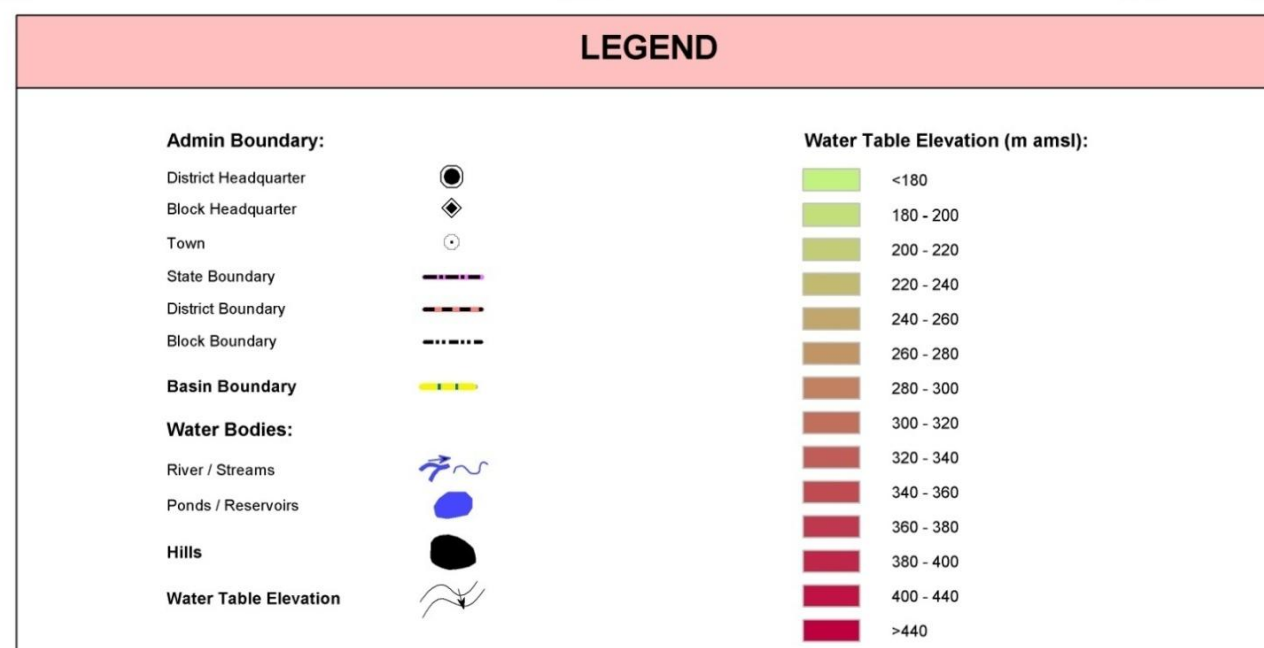
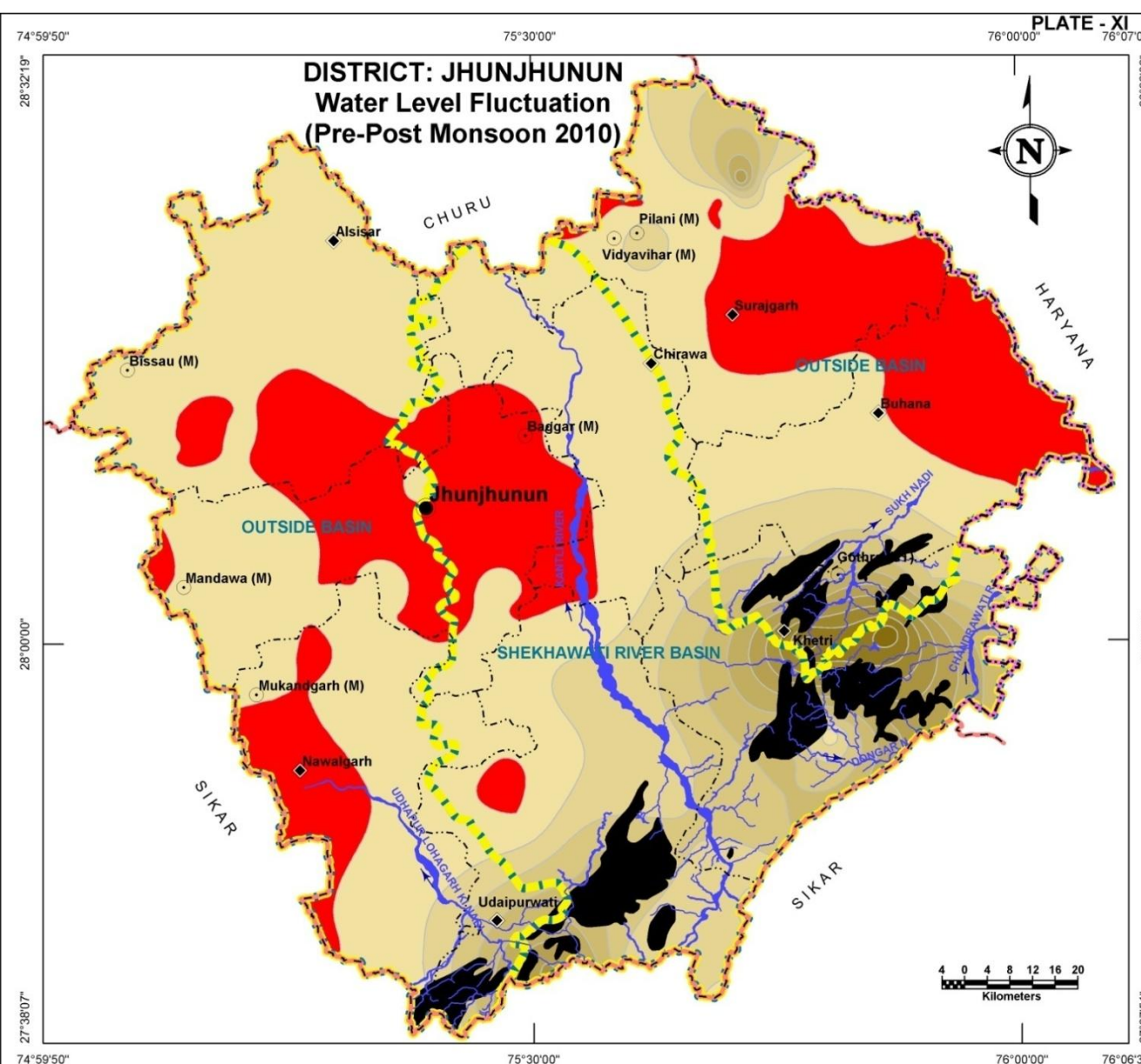
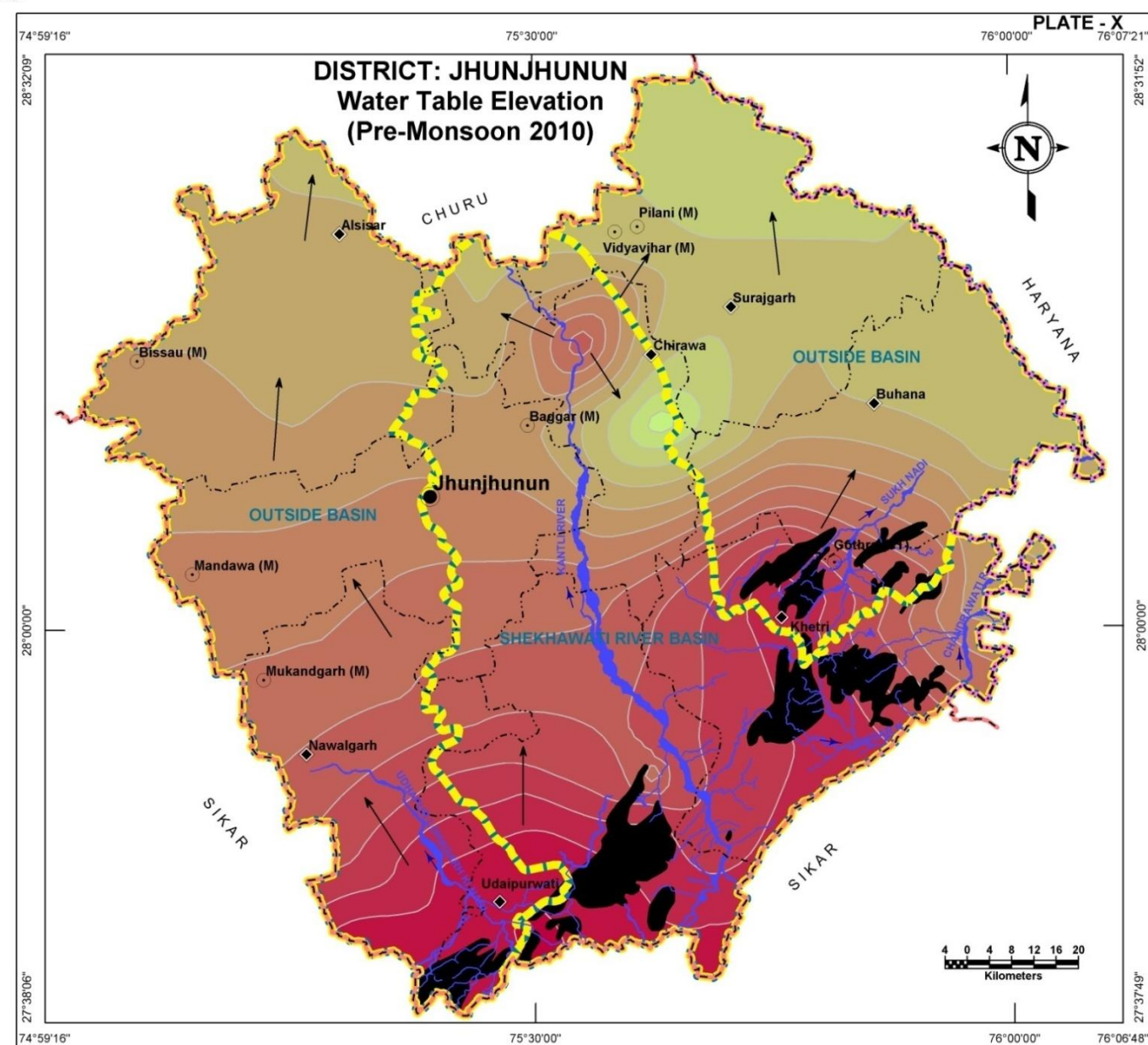
Block Name	Block wise area coverage (sq km) per water elevation range (m amsl)														Total Area (sq km)
	< 180	180 - 200	200 - 220	220 - 240	240 - 260	260 - 280	280 - 300	300 - 320	320 - 340	340 - 360	360 - 380	380 - 400	400 - 440	> 440	
Alsisar	-	-	-	38.2	465.8	314.9	-	-	-	-	-	-	-	-	818.9
Buhana	-	-	3.8	281.3	118.4	80.6	66.8	54.3	30.0	6.2	-	-	-	-	641.4
Chirawa	4.3	22.5	39.5	74.2	123.6	96.8	87.6	56.0	17.4	-	-	-	-	-	521.9
Jhunjhunun	-	-	-	-	39.4	242.2	362.8	101.4	18.5	-	-	-	-	-	764.3
Khetri	-	-	-	-	4.2	36.7	34.0	49.8	74.0	123.3	228.2	124.9	1.3	-	676.4
Nawalgarh	-	-	-	-	-	-	36.8	194.2	133.2	86.3	50.6	65.5	69.2	-	635.8
Surajgarh	-	7.2	312.7	413.6	74.9	6.0	1.1	-	-	-	-	-	-	-	815.5
Udaipurwati	-	-	-	-	-	-	-	52.4	164.7	155.4	94.0	105.2	172.9	14.5	759.1
(sq km)	4.3	29.7	356.0	807.3	826.3	777.2	589.1	508.1	437.8	371.2	372.8	295.6	243.4	14.5	5,633.3

WATER LEVEL FLUCTUATION (PRE TO POST MONSOON 2010)

A 2m contour interval adopted to visualize the ground water level fluctuation reveals the fluctuation to range from upto -2m to more than 18m, as seen in Plate– XI. The negative fluctuation areas (<0m, indicated red regions) are the areas where water level has gone down in post monsoon season with respect to pre- monsoon water levels. Such large ground water depletion areas are located around Jhunjhunun, Nawalgarh, and Surajgarh. Rest of the district has shown a general to significant rise in ground water level in the post monsoon season with respect to pre monsoon season with maximum rise of more than 18m is noticed in eastern part of Khekri.

Table: Block wise area covered in each water fluctuation zone

Water level fluctuation range (m)	Block wise area coverage (sq km)								Total Area (sq km)
	Alsisar	Buhana	Chirawa	Jhunjhunun	Khetri	Nawalgarh	Surajgarh	Udaipurwati	
< 0	187.4	276.5	42.4	406.9	-	177.4	305.5	32.9	1,429.0
0 to 2	631.5	241.6	464.1	357.4	7.0	427.3	373.9	232.2	2,735.0
2 to 4	-	103.0	15.4	-	111.2	29.3	72.6	306.8	638.3
4 to 6	-	18.1	-	-	199.8	1.8	40.6	110.4	370.7
6 to 8	-	2.2	-	-	126.0	-	16.5	44.5	189.2
8 to 10	-	-	-	-	75.5	-	5.1	21.1	101.7
10 to 12	-	-	-	-	53.3	-	1.3	9.2	63.8
12 to 14	-	-	-	-	32.8	-	-	2.0	34.8
14 to 16	-	-	-	-	28.5	-	-	-	28.5
16 to 18	-	-	-	-	36.8	-	-	-	36.8
> 18	-	-	-	-	5.5	-	-	-	5.5
Total	818.9	641.4	521.9	764.3	676.4	635.8	815.5	759.1	5,633.3



GROUND WATER ELECTRICAL CONDUCTIVITY DISTRIBUTION

DISTRICT – JHUNJHUNUN

The Electrical conductivity (at 25°C) distribution map is presented in Plate – XII. The areas with low EC values in ground water (<2000 µS/cm) are shown in yellow color which occupy almost 73% of the district area indicating that, by and large the ground water in is suitable for domestic purpose. The areas with moderately high EC values (2000 -4000 µS/cm) are shown in green color occupy 20% of the district area, largely northwestern part of the district and around Buhana. Remaining small part of the district (approximately 7%) has high EC values in ground water (>4000 µS/cm, shown in red color) area largely present in northwestern part of the district where the ground water is not suitable for domestic purpose.

Table: Block wise area of Electrical conductivity distribution

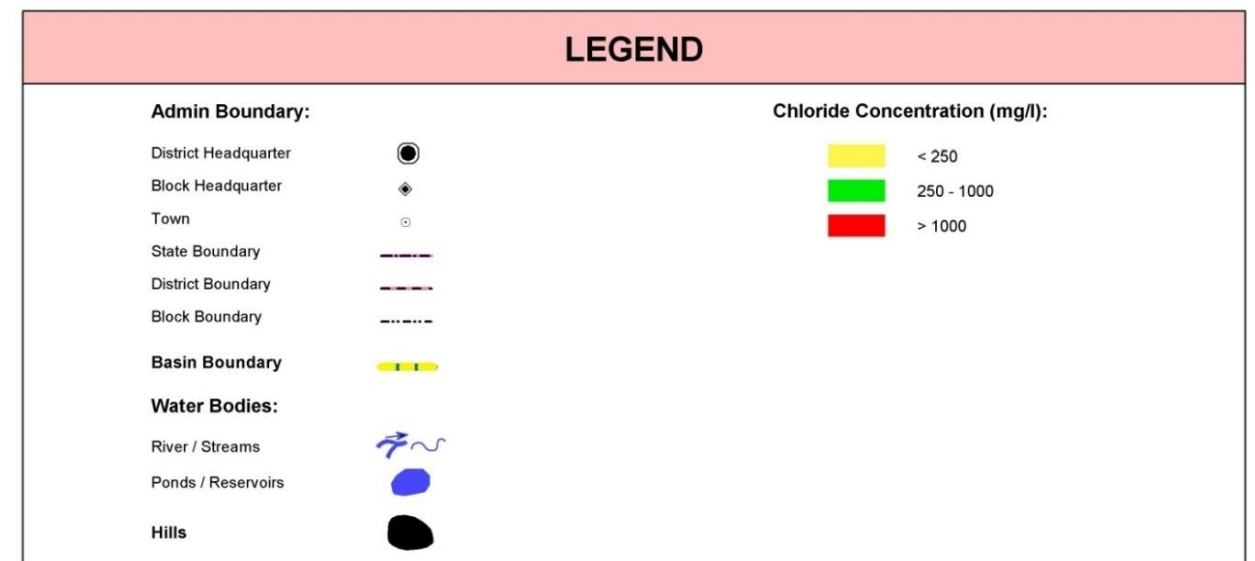
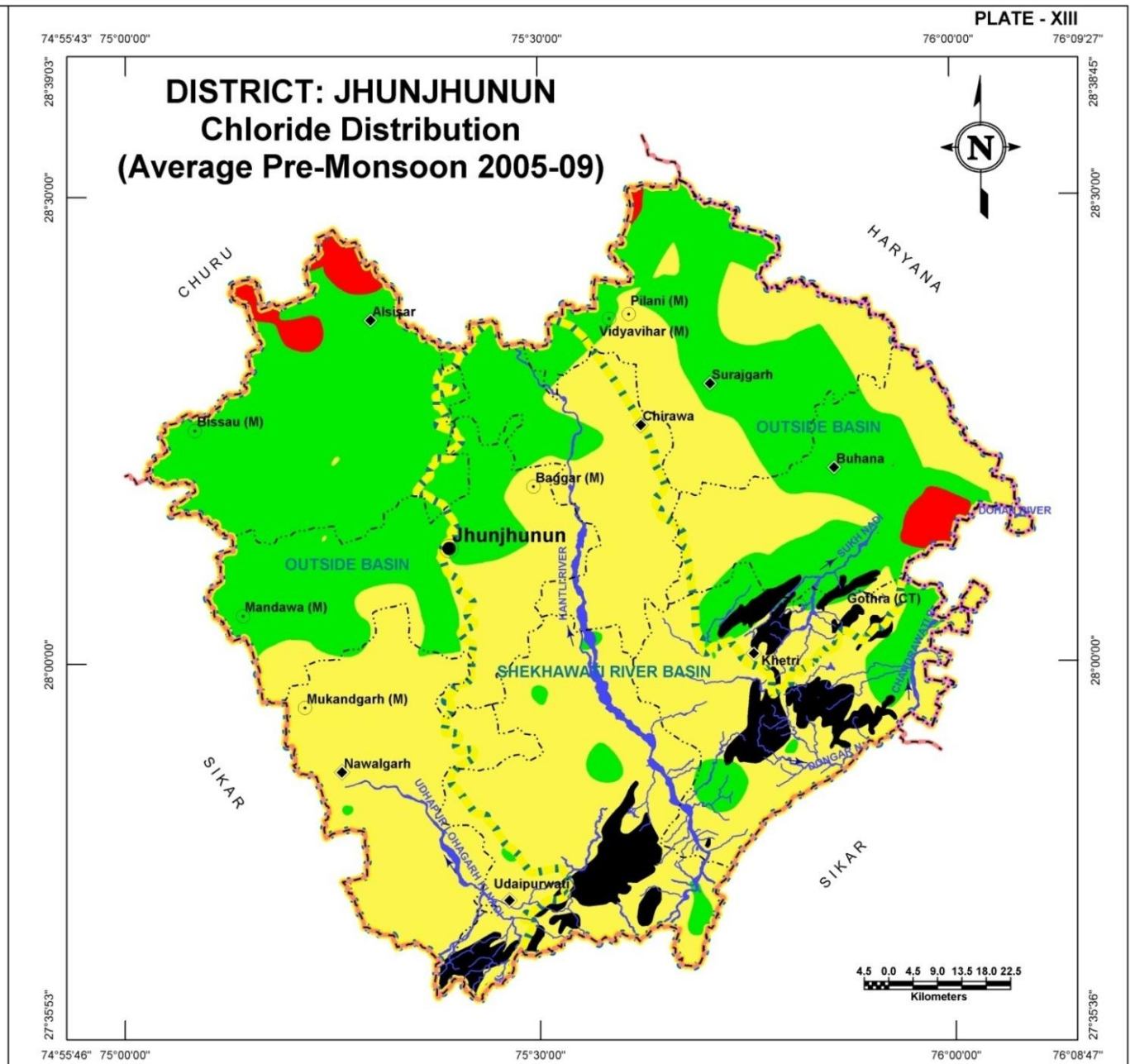
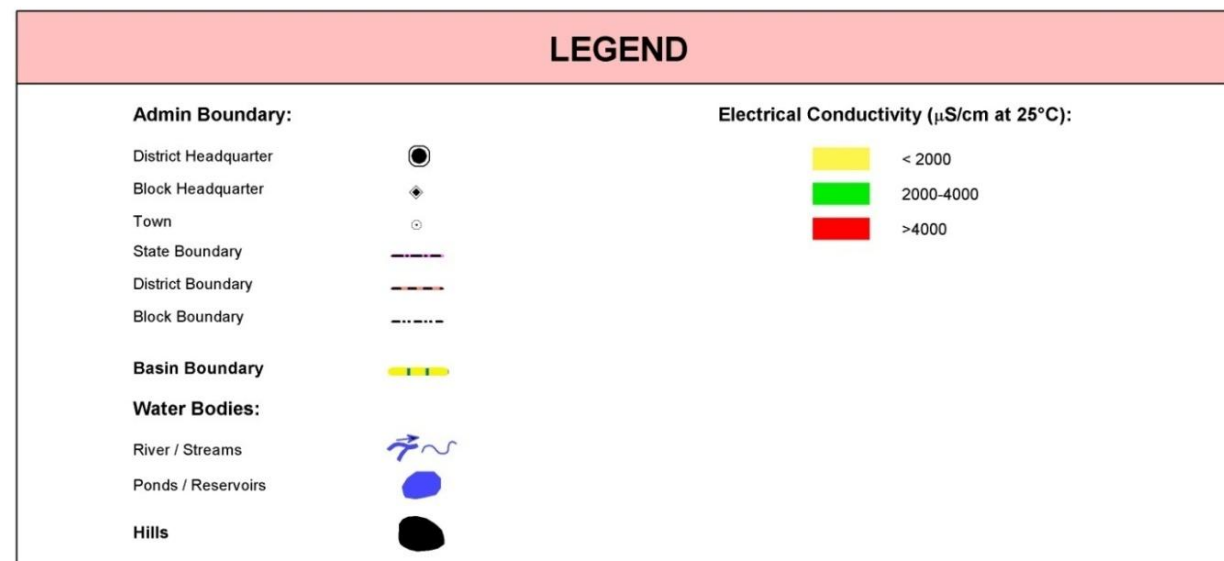
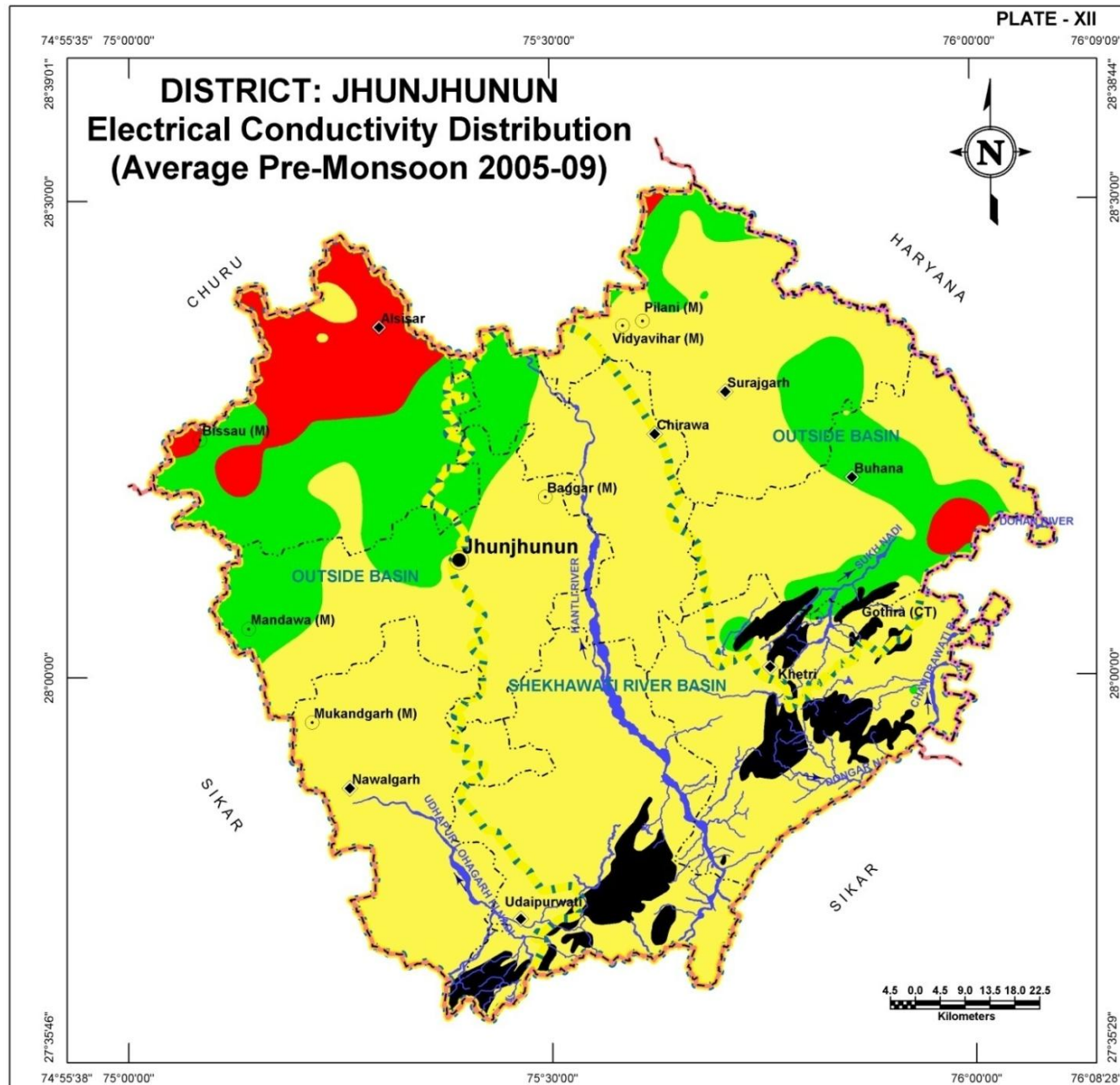
Electrical Conductivity Ranges (µS/cm at 25°C) (Ave. of years 2005-09)	Block wise area coverage (sq km)																Total Area (sq km)
	Alsisar		Buhana		Chirawa		Jhunjhunun		Khetri		Nawalgarh		Surajgarh		Udaipurwati		
	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	
< 2000	57.8	7.0	404.7	63.1	434.2	83.2	541.3	70.8	660.0	97.6	635.8	100.0	626.5	76.8	759.1	100.0	4,119.4
2000-4000	423.7	51.8	202.7	31.6	87.7	16.8	223.0	29.2	16.4	2.4	-	-	183.0	22.5	-	-	1,136.5
>4000	337.4	41.2	34.0	5.3	-	-	-	-	-	-	-	-	6.0	0.7	-	-	377.4
Total	818.9	100.0	641.4	100.0	521.9	100.0	764.3	100.0	676.4	100.0	635.8	100.0	815.5	100.0	759.1	100.0	5,633.3

GROUND WATER CHLORIDE DISTRIBUTION

High chloride concentration in ground water also renders it unsuitable for domestic and other purposes. The yellow colored regions in Plate – XIII are such areas where chloride concentration is low (<250 mg/l) and these areas occupy approximately 57% of the district area and is suitable for domestic purpose. The areas with moderately high chloride concentration (250-1000mg/l) are shown in green color occupying about 41% of the district area. Remaining small part of the district approximately 2% falls under high chloride concentration (>1000 mg/l) category, largely northwestern part of Alsisar where the ground water is not suitable for domestic purpose.

Table: Block wise area of Chloride distribution

Chloride Concentration Range(mg/l) (Ave. of years 2005-09)	Block wise area coverage (sq km)																Total Area (sq km)
	Alsisar		Buhana		Chirawa		Jhunjhunun		Khetri		Nawalgarh		Surajgarh		Udaipurwati		
	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	
< 250	0.6	-	300.8	47.0	366.2	70.0	310.4	41.0	524.1	77.0	631.6	99.0	336.1	41.0	720.4	95.0	3,190.2
250-1000	746.4	91.0	298.0	46.0	155.7	30.0	453.9	59.0	152.3	23.0	4.2	1.0	472.6	58.0	38.7	5.0	2,321.8
> 1000	71.9	9.0	42.6	7.0	-	-	-	-	-	-	-	-	6.8	1.0	-	-	121.3
Total	818.9	100.0	641.4	100.0	521.9	100.0	764.3	100.0	676.4	100.0	635.8	100.0	815.5	100.0	759.1	100.0	5,633.3



GROUND WATER FLUORIDE DISTRIBUTION

The Fluoride concentration map is presented in Plate – XIV. The areas with low concentration (i.e., >1.5 mg/l) are shown in yellow color and such areas occupy almost 69% of the district area which is suitable for domestic purpose. The areas with moderately high concentration (1.5-3.0 mg/l) are shown in green color and occupying approximately 25% of the district area, largely around Mandawa, Bissau and Baggar. Remaining small part of the district (approximately 6%) has high Fluoride concentration (>3.0 mg/l, shown in red color), are largely seen around Alsisar. The ground water in this region is not suitable for domestic purpose.

Table: Block wise area of Fluoride distribution

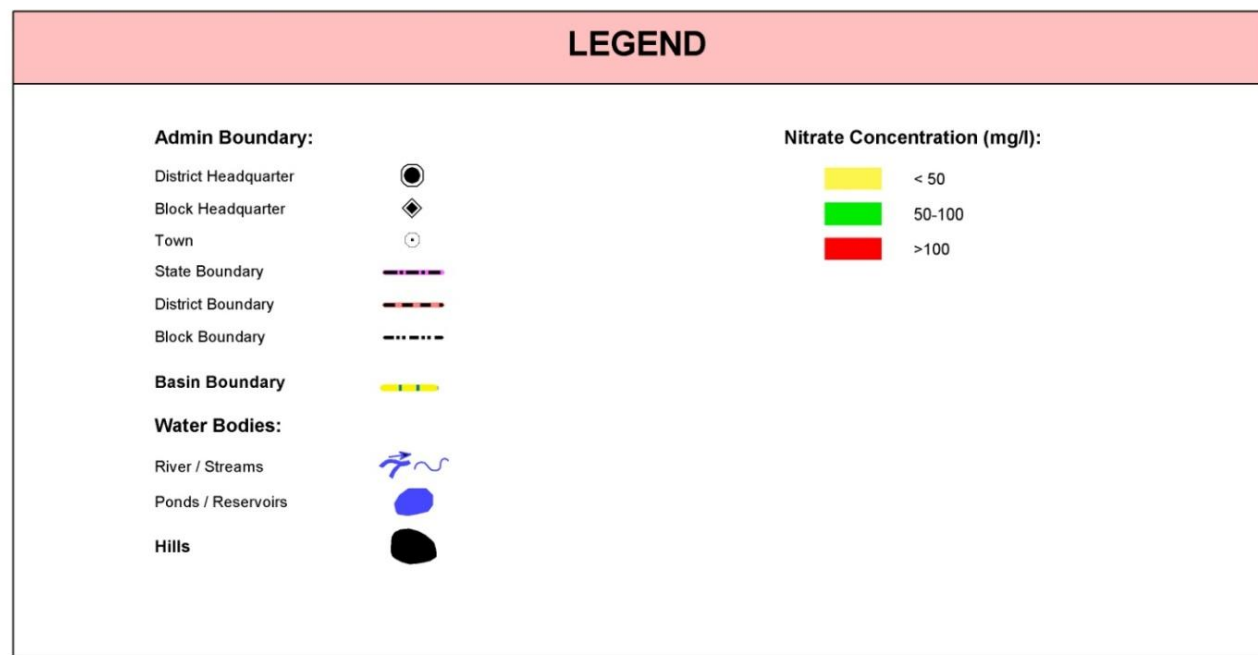
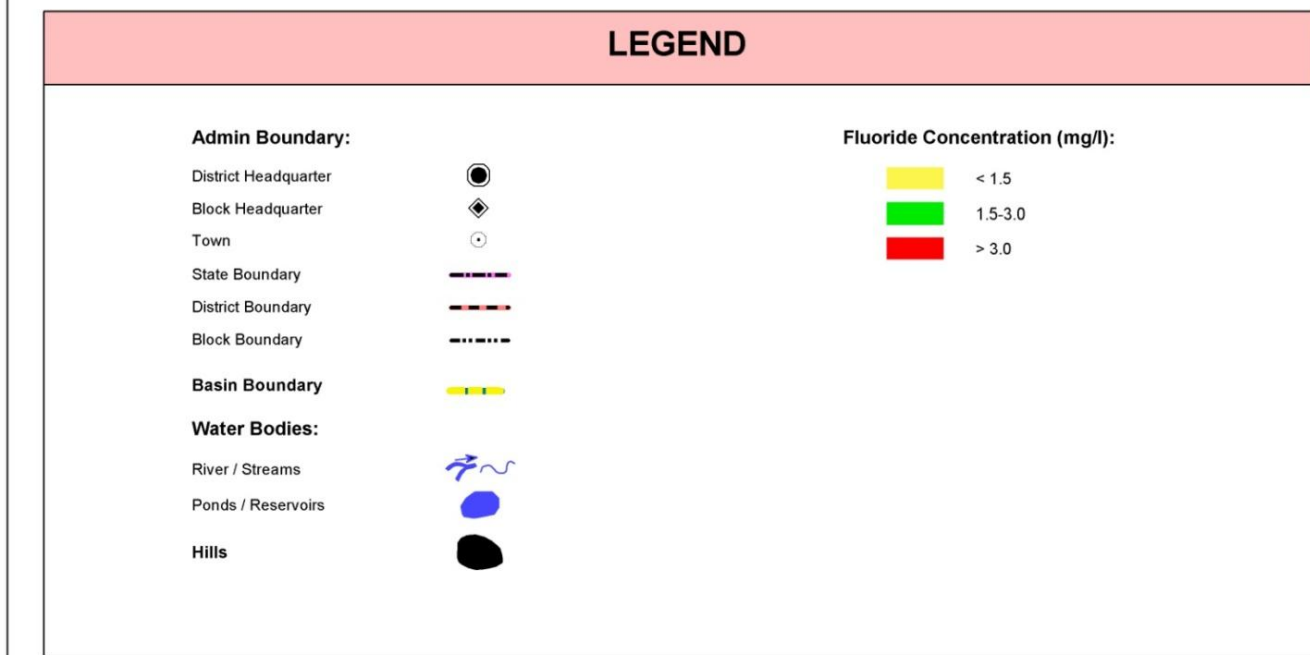
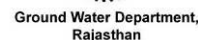
Fluoride concentration Range(mg/l) (Ave. of years 2005-09)	Block wise area coverage (sq km)																Total Area (sq km)
	Alsisar		Buhana		Chirawa		Jhunjhunun		Khetri		Nawalgarh		Surajgarh		Udaipurwati		
	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	
< 1.5	46.6	5.7	504.6	78.7	319.2	61.2	302.8	39.6	665.6	98.4	627.9	98.8	755.0	92.6	675.8	89.0	3,897.5
1.5-3.0	501.0	61.2	117.8	18.3	172.5	33.0	455.2	59.6	10.8	1.6	7.9	1.2	57.6	7.0	83.3	11.0	1,406.1
> 3.0	271.3	33.1	19.0	3.0	30.2	5.8	6.3	0.8	-	-	-	-	2.9	0.4	-	-	329.7
Total	818.9	100.0	641.4	100.0	521.9	100.0	764.3	100.0	676.4	100.0	635.8	100.0	815.5	100.0	759.1	100.0	5,633.3

GROUND WATER NITRATE DISTRIBUTION

High nitrate concentration in ground water renders it unsuitable for agriculture purposes. Plate – XV shows distribution of Nitrate in ground water. Low nitrate concentration (<50 mg/l) is shown in yellow color and occupying about 36% of the district area which is suitable for agriculture purpose. The areas with moderately high nitrate concentration (50-100 mg/l) are shown in green colors which occupy approximately 41% of the district area. Remaining part of the district area (approximately 20%) is covered with high nitrate concentration (>100 mg/l) which is shown in red colored patches where the ground water in this not suitable for agriculture purpose.

Table: Block wise area of Nitrate distribution

Nitrate concentration Range(mg/l) (Ave. of years 2005-09)	Block wise area coverage (sq km)																Total Area (sq km)
	Alsisar		Buhana		Chirawa		Jhunjhunun		Khetri		Nawalgarh		Surajgarh		Udaipurwati		
	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	
< 50	8.4	1.0	322.9	50.3	252.3	48.3	134.2	17.6	456.8	67.5	231.6	36.4	80.0	9.8	558.2	73.5	2,044.4
50-100	162.7	19.9	161.1	25.1	233.0	44.7	441.7	57.8	110.0	16.3	379.9	59.8	663.1	81.3	148.8	19.6	2,300.3
>100	647.8	79.1	157.4	24.6	36.6	7.0	188.4	24.6	109.6	16.2	24.3	3.8	72.4	8.9	52.1	6.9	1,288.6
Total	818.9	100.0	641.4	100.0	521.9	100.0	764.3	100.0	676.4	100.0	635.8	100.0	815.5	100.0	759.1	100.0	5,633.3



DEPTH TO BEDROCK

DISTRICT – JHUNJHUNUN

The entire area of the district is underlain by the hardrocks at different depths. From hydrogeological perspective, the beginning of massive bedrock has been considered for defining top of bedrock surface. Depth to bedrock map of Jhunjhunun district (Plate – XVI) reveals wide variation of more than 80m reaching a maximum depth of more than 120m bgl in the eastern, western and northern part of the district. Areas around Alsisar, Buhana, Chirawa, Jhunjhunun, Nawalgarh, Surajgarh and Udaipurwati blocks indicate the occurrence of bedrock at very deep levels. In the central part of the district, moderately deep bedrock depths are encountered. Shallow bedrock depth is mostly found in eastern part of Khetri and Udaipurwati block. Deepest occurrence of bedrock (indicating high alluvial thickness) is found in eastern part of Nawalgarh block i.e., more than 120 m bgl.

Depth to bedrock (m bgl)	Block wise area coverage (sq km)																Total Area (sq km)
	Alsisar		Buhana		Chirawa		Jhunjhunun		Khetri		Nawalgarh		Surajgarh		Udaipurwati		
	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	
< 40	-	-	-	-	-	-	-	-	65.7	10.0	-	-	-	-	45.0	6.0	110.7
40-60	-	-	21.9	3.0	-	-	-	-	326.1	48.0	-	-	-	-	104.5	14.0	452.5
60-80	21.1	3.0	247.5	39.0	24.6	4.7	2.3	0.3	150.8	22.0	50.0	8.0	-	-	237.2	31.0	733.5
80-100	596.6	73.0	362.1	56.0	458.9	87.9	585.5	76.6	133.8	20.0	130.1	20.0	277.5	34.0	289.7	38.0	2,834.2
100-120	201.2	24.0	9.9	2.0	38.4	7.4	176.5	23.1	-	-	432.7	68.0	538.0	66.0	82.7	11.0	1,479.4
> 120	-	-	-	-	-	-	-	-	-	-	23.0	4.0	-	-	-	-	23.0
Total	818.9	100.0	641.4	100.0	521.9	100.0	764.3	100.0	676.4	100.0	635.8	100.0	815.5	100.0	759.1	100.0	5,633.3

UNCONFINED AQUIFER

Alluvial areas:

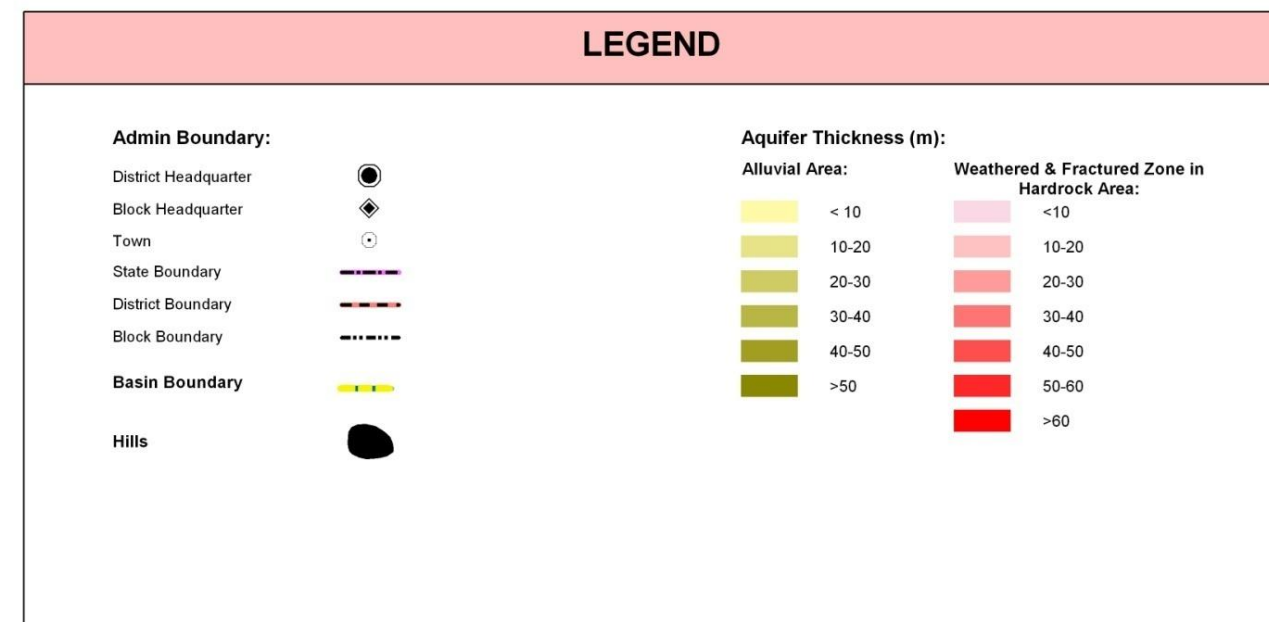
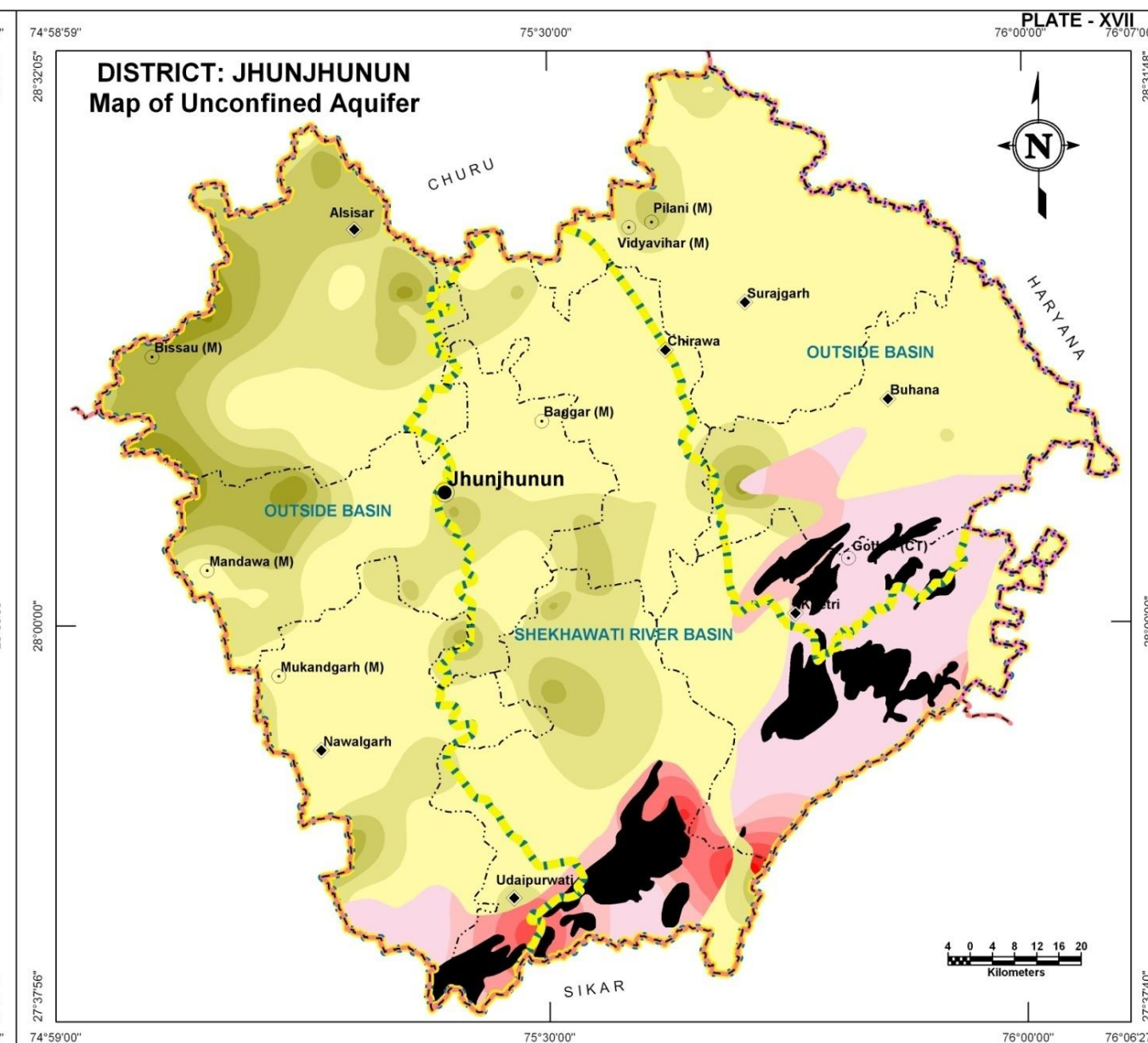
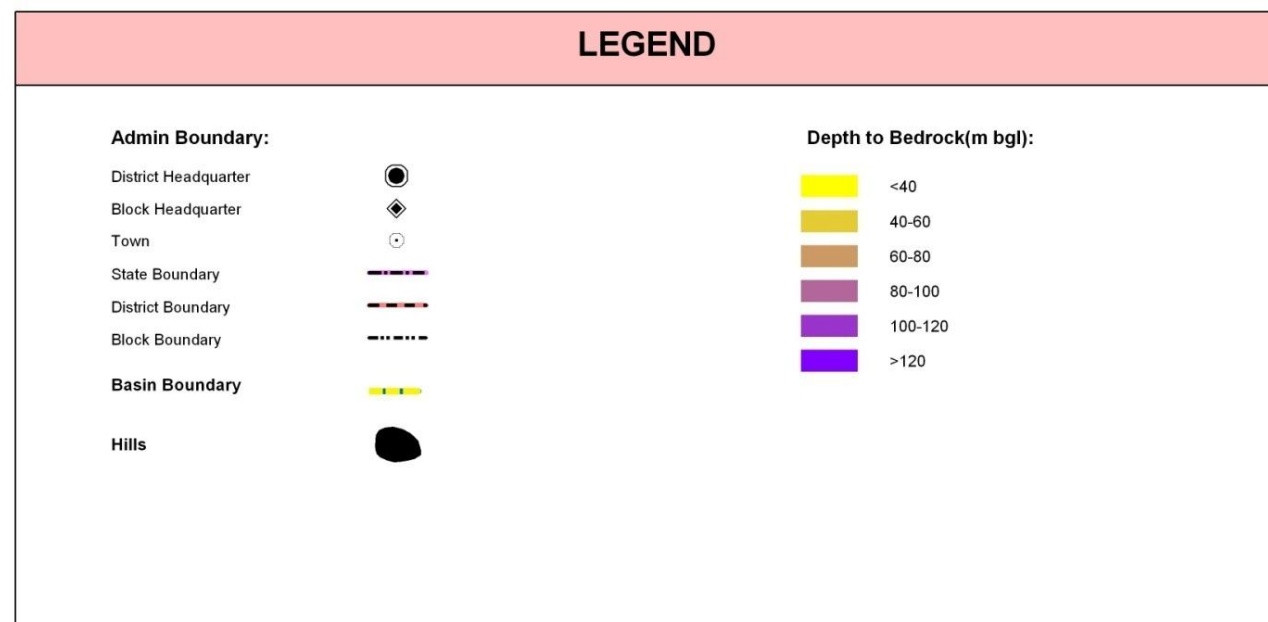
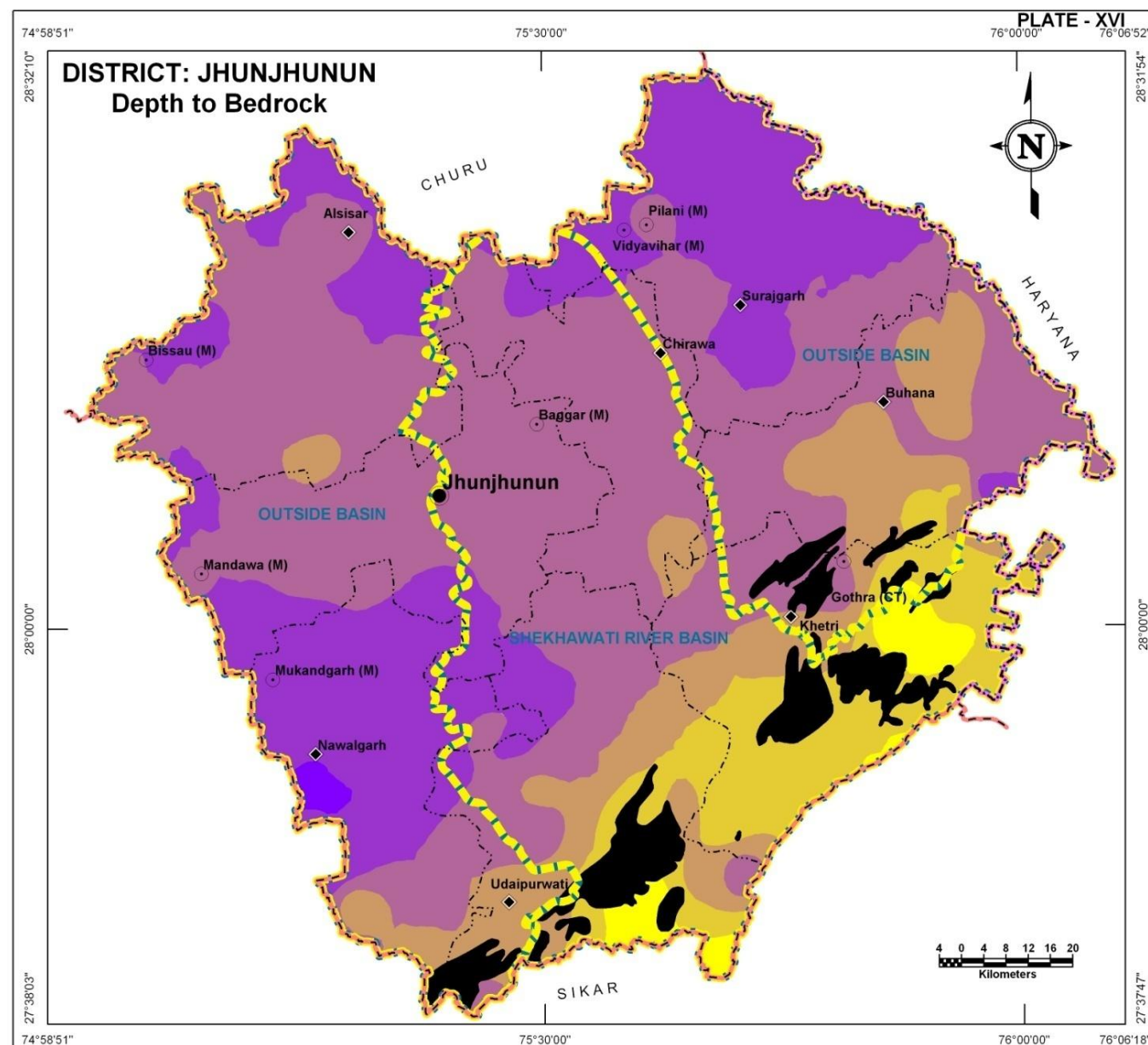
Most part of the district excluding some eastern and southern parts has thick cover of alluvium both younger and older. In unconfined conditions the alluvial aquifer attains a thickness of more than 50m. Perusal of Plate – XVII reveals a moderate thickness of upto 30m in the area with pockets of more thick (>50m) unconfined alluvial aquifer material in the western fringe of Alsiar block.

Hard rock areas:

Weathered, fractured and jointed rock formations occurring at shallower depths constitute good unconfined aquifers. Such aquifers under unconfined conditions are located in the southeastern parts of the district encompassing the Buhana, Khetri, Nawalgarh and Udaipurwati blocks. This productive zone attains very high thickness of more than 50m to the south of Udaipurwati block.

Unconfined aquifer Thickness (m)	Block wise area coverage(sq km)								Total Area (sq km)
	Alsisar	Buhana	Chirawa	Jhunjhunun	Khetri	Nawalgarh	Surajgarh	Udaipurwati	
< 10	185.4	430.1	395.0	395.2	251.2	434.8	730.9	292.3	3,114.9
10-20	217.8	42.8	106.3	226.6	9.3	105.0	74.6	223.6	1,006.0
20-30	241.2	21.8	20.6	63.7	-	23.0	10.0	55.9	436.2
30-40	134.7	6.9	-	63.6	-	6.8	-	5.6	217.6
40-50	38.2	0.6	-	15.2	-	-	-	-	54.0
> 50	1.6	-	-	-	-	-	-	-	1.6
Total	818.9	502.2	521.9	764.3	260.5	569.6	815.5	577.4	4,830.3

Unconfined aquifer Thickness (m)	Block wise area coverage(sq km)								Total Area (sq km)
	Alsisar	Buhana	Chirawa	Jhunjhunun	Khetri	Nawalgarh	Surajgarh	Udaipurwati	
<10	-	97.4	-	-	328.8	56.8	-	28.4	511.4
10-20	-	41.8	-	-	60.7	9.2	-	36.3	148.0
20-30	-	-	-	-	11.5	0.2	-	43.2	54.9
30-40	-	-	-	-	10.4	-	-	50.1	60.5
40-50	-	-	-	-	4.0	-	-	19.3	23.3
50-60	-	-	-	-	0.5	-	-	3.5	4.0
> 60	-	-	-	-	-	-	-	0.9	0.9
Total	-	139.2	-	-	415.9	66.2	-	181.7	803.0



Glossary of terms

S. No.	Technical Terms	Definition
1	AQUIFER	A saturated geological formation which has good permeability to supply sufficient quantity of water to a Tube well, well or spring.
2	ARID CLIMATE	Climate characterized by high evaporation and low precipitation.
3	ARTIFICIAL RECHARGE	Addition of water to a ground water reservoir by man-made activity
4	CLIMATE	The sum total of all atmospheric or meteorological influences principally temperature, moisture, wind, pressure and evaporation of a region.
5	CONFINED AQUIFER	A water bearing strata having confined impermeable overburden. In this aquifer, water level represents the piezometric head.
6	CONTAMINATION	Introduction of undesirable substance, normally not found in water, which renders the water unfit for its intended use.
7	DRAWDOWN	The drawdown is the depth by which water level is lowered.
8	FRESH WATER	Water suitable for drinking purpose.
9	GROUND WATER	Water found below the land surface.
10	GROUND WATER BASIN	A hydro-geologic unit containing one large aquifer or several connected and interrelated aquifers.
11	GROUND WATER RECHARGE	The natural infiltration of surface water into the ground.
12	HARD WATER	The water which does not produce sufficient foam with soap.
13	HYDRAULIC CONDUCTIVITY	A constant that serves as a measure of permeability of porous medium.
14	HYDROGEOLOGY	The science related with the ground water.
15	HUMID CLIMATE	The area having high moisture content.
16	ISOHYET	A line of equal amount of rainfall.
17	METEOROLOGY	Science of the atmosphere.
18	PERCOLATION	It is flow through a porous substance.
19	PERMEABILITY	The property or capacity of a soil or rock for transmitting water.
20	pH	Value of hydrogen-ion concentration in water. Used as an indicator of acidity (pH < 7) or alkalinity (pH > 7).
21	PIEZOMETRIC HEAD	Elevation to which water will rise in a piezometers.
22	RECHARGE	It is a natural or artificial process by which water is added from outside to the aquifer.
23	SAFE YIELD	Amount of water which can be extracted from ground water without producing undesirable effect.
24	SALINITY	Concentration of dissolved salts.
25	SEMI-ARID	An area is considered semiarid having annual rainfall between 10-20 inches.
26	SEMI-CONFINED AQUIFER	Aquifer overlain and/or underlain by a relatively thin semi-pervious layer.
27	SPECIFIC YIELD	Quantity of water which is released by a formation after it's complete saturation.
28	TOTAL DISSOLVED SOLIDS	Total weight of dissolved mineral constituents in water per unit volume (or weight) of water in the sample.

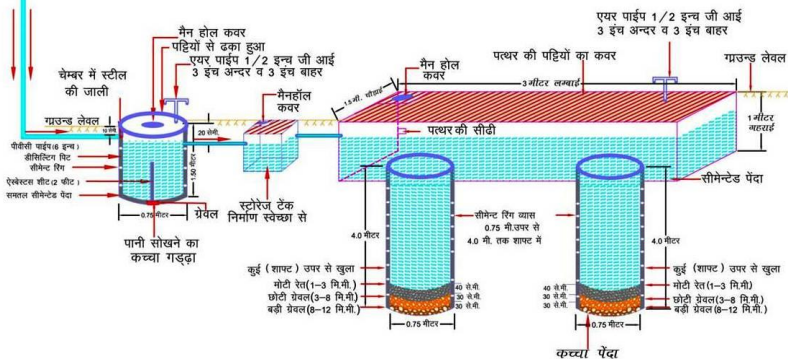
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S. No.	Technical Terms	Definition
29	TRANSMISSIBILITY	It is defined as the rate of flow through an aquifer of unit width and total saturation depth under unit hydraulic gradient. It is equal to product of full saturation depth of aquifer and its coefficient of permeability.
30	UNCONFINED AQUIFER	A water bearing formation having permeable overburden. The water table forms the upper boundary of the aquifer.
31	UNSATURATED ZONE	The zone below the land surface in which pore space contains both water and air.
32	WATER CONSERVATION	Optimal use and proper storage of water.
33	WATER RESOURCES	Availability of surface and ground water.
34	WATER RESOURCES MANAGEMENT	Planned development, distribution and use of water resources.
35	WATER TABLE	Water table is the upper surface of the zone of saturation at atmospheric pressure.
36	ZONE OF SATURATION	The ground in which all pores are completely filled with water.
37	ELECTRICAL CONDUCTIVITY	Flow of free ions in the water at 25C mu/cm.
38	CROSS SECTION	A Vertical Projection showing sub-surface formations encountered in a specific plane.
39	3-D PICTURE	A structure showing all three dimensions i.e. length, width and depth.
40	GWD	Ground Water Department
41	CGWB	Central Ground Water Board
42	CGWA	Central Ground Water Authority
43	SWRPD	State Water Resources Planning Department
44	EU-SPP	European Union State Partnership Programme
45	TOPOGRAPHY	Details of drainage lines and physical features of land surface on a map.
46	GEOLOGY	The science related with the Earth.
47	GEOMORPHOLOGY	The description and interpretation of land forms.
48	PRE MONSOON SURVEY	Monitoring of Ground Water level from the selected DKW/Piezometer before Monsoon (carried out between 15th May to 15th June)
49	POST-MONSOON SURVEY	Monitoring of Ground Water level from the selected DKW/Piezometer after Monsoon (carried out between 15th October to 15th November)
50	PIEZOMETER	A non-pumping small diameter bore hole used for monitoring of static water level.
51	GROUND WATER FLUCTUATION	Change in static water level below ground level.
52	WATER TABLE	The static water level found in unconfined aquifer.
53	DEPTH OF BED ROCK	Hard & compact rock encountered below land Surface.
54	G.W. MONITORING STATION	Dug wells selected on grid basis for monitoring of state water level.
55	EOLIAN DEPOSITS	Wind-blown sand deposits



भवन छत क्षेत्रफल 300 से 500 वर्गमीटर तक
निर्माण किये जाने वाले मुख्य भाग एवं डिजाइन

- PVC पाईप 6" व्यास
- सीमेंट रिंग से निर्मित डीसिल्टिंग पिट
(0.75 मी व्यास x 1.50 मी गहरा)
- रीचारज टैंक 1.5 मी. चौड़ा x 3 मी. लम्बा x 1 मी. गहरा
- सीमेंट रिंग से निर्मित शाफ्ट
(0.75 मी व्यास x 4 मी. गहरा) (संख्या 2)
- संरचना की अनुमानित लागत रु 24,000 अधिक
- वार्षिक पुनर्भरित जल लगभग 2,00,000 लीटर
- 20 वर्षों में पुनर्भरित जल लगभग 40,00,000 लीटर
- पुनर्भरित जल की लागत 1 पैसे प्रति लीटर से कम



चित्र-4

भूजल में घुले मुख्य तत्वों की अधिकता का मानव शरीर पर दुष्प्रभाव



बोरोन-स्नायु तन्त्र पर प्रभाव

फ्लोराइड - दंत क्षरण

क्लोराइड-सोडियम के साथ
मिलकर उच्च रक्त चाप

सोडियम-हृदय, गुर्दा व रक्त
परिसंचरण रोगों से
ग्रसित लोगों को हानिकारक

कैल्शियम-जोड़ों में कड़ापन

नाइट्रेट-नवजात शिशुओं
ब्लू बेबी बीमारी
(मेथमोग्लोबिनिमिया)

आर्सेनिक-त्वचा रोग, कैंसर

सल्फेट-अधिकता में मैग्नेशियम के साथ मिलकर दस्तावर

लेड-बच्चों के शारीरिक
व मानसिक विकास में बाधा
वयस्कों में गुर्दे के रोग

आयरन-आयरन जीवाणु से आमाशय संबंधी रोग

फ्लोराइड-जोड़ों में अकड़न,
हड्डियों में मृदाव



केन्द्रीय भूमि जल बोर्ड,
पश्चिमी क्षेत्र, जयपुर
जल संसाधन मंत्रालय
भारत सरकार
-mail:cgwbwr@sancharnet.in

भूजल अमूल्य है इसे प्रदूषित न करें।

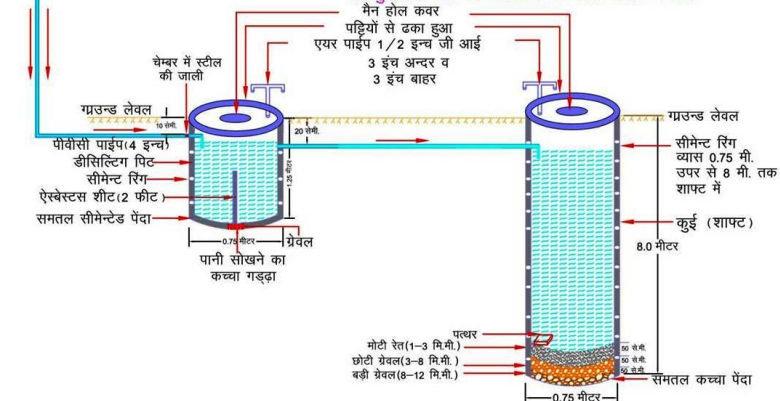


भवन छत क्षेत्रफल 100 से 200 वर्गमीटर तक

चित्र-2

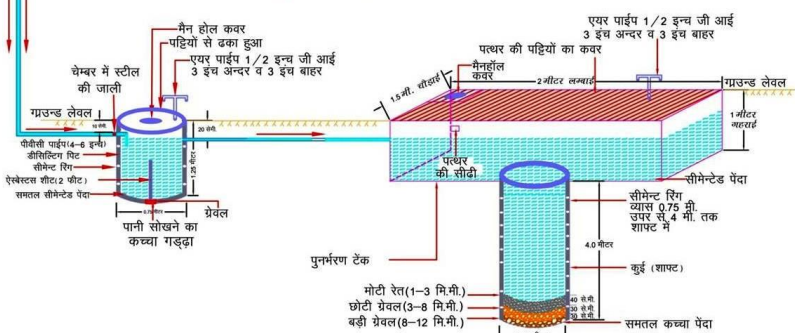
निर्माण किये जाने वाले मुख्य भाग एवं डिज़ाइन

- PVC पाईप 4" व्यास
- सोनेन्ट रिंग से निर्मित डीसिल्टिंग पिट
(0.75 मी व्यास x 125 मी. गहरा)
- सोनेन्ट रिंग से निर्मित शाफ्ट
(0.75 मी व्यास x 4 मी गहरी)
- संरचना की अनुमानित लागत रु 11,000–12,000
- वार्षिक पुनर्मरिज जल लगभग 83,000 लीटर
- 20 वर्षों में पुनर्मरिज जल लगभग 16,64,000 लीटर
- पुनर्मरिज जल की लागत 1 पैसे प्रति लीटर से कम



भवन छत क्षेत्रफल 200 से 300 वर्गमीटर तक निर्माण किये जाने वाले मुख्य भाग एवं डिजाईन

- PVC पाईप 4" – 6" व्यास
- सीमेन्ट रिंग से निर्मित डीसिल्टिंग पिट
(0.75 मी व्यासx1.25 मी गहरा)
- रोवाज़ टैंक 1.5 मी.घोड़ाx2 मी.लम्बाx1 मी.गहरा
- सीमेन्ट रिंग से निर्मित शाफ्ट
(0.75 मी व्यासx4 मी.गहरी)
- संरचना की अनुमानित लागत रु 15,000–16,000
- वार्षिक पुनर्भरित जल लगभग 1,25,000 लीटर
- 20 वर्षों में पुनर्भरित जल लगभग 25,00,000 लीटर
- पुनर्भरित जल की लागत 1 पैसे प्रति लीटर से कम



चित्र-3

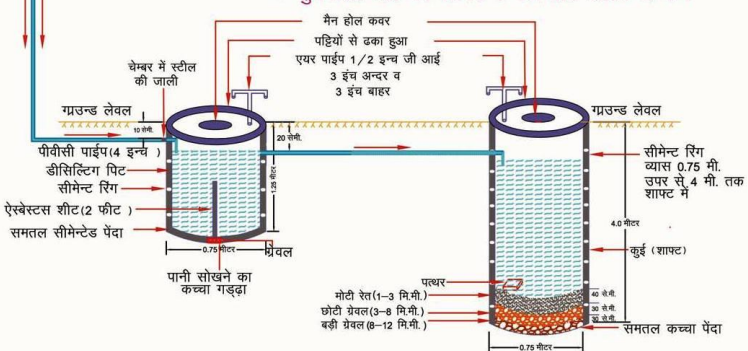


भवन छत क्षेत्रफल 100 वर्गमीटर तक

चित्र-1

निर्माण किये जाने वाले मुख्य भाग एवं डिज़ाइन

- PVC पाईप 4" व्यास
- सीमेन्ट रिंग से निर्मित डीसिल्टिंग पिट
(0.75 मी व्यासx 1.25 मी गहरा)
- सीमेन्ट रिंग से निर्मित शाफ्ट
(0.75 मी व्यासx 4 मी गहरी)
- संरचना की अनुमानित लागत रु 7000–8000
- वार्षिक पुनर्भरित जल लगभग 40,000 लीटर
- 20 वर्षों में पुनर्भरित जल लगभग 8,00,000 लीटर
- पुनर्भरित जल की लागत 1 पैसे प्रति लीटर से कम



Myths and Facts about Ground Water

S No	Myths	Facts
1	What is Ground Water • an underground lake • a net work of underground rivers • a bowl filled with water	Water which occurs below the land in geological formations/rocks is Ground water
2	Ground Water occurs everywhere beneath the Land Surface	Not really, it depends on the nature of rock formation
3	There is a relationship between ground water and surface water	Not all the places. Near streams/rivers there is relation
4	Groundwater is not renewable resource	It is renewable source and every year it is being recharged through rain/applied irrigation etc
5	Ground water is unlimited and deeper you drill more discharge	It is limited to annual recharge from rain/applied irrigation. The discharge may not increase if you go deeper
6	Ground Water moves rapidly	The movement of ground water is very slow
7	Ground water pumped from wells is thousands of years old	Generally the ground water being tapped through wells is a few years old
8	If water taste good—it is safe to drink	It may have other chemicals e.g. fluoride, nitrates etc which are harmful
9	Water from free flowing tube wells is very pure	This water can also be contaminated so test before use
10	If I recharge my TW/DW/HP it will not benefit me	It will also benefit you and also adjoining wells
11	There is no static ground water resources in Rajasthan	Rajasthan is also having Static GW resources, and being tapped in most of areas as GW annual withdrawal is more than annual recharge
12	I cannot meet annual cooking and drinking water requirement by rain water harvesting	The water requirement for drinking and cooking is only 8 lit/day. You can harvest this water for family of 5 persons from roof top or paved area of 75 Sq m to meet annual requirement
13	You can increase ground water recharge	This can be done by harvesting the rain water and storing in sub surface reservoir (GW) by constructing the recharge structures
14	You cannot use abandoned TW/HP/DW for ground water recharge	These should be used as recharge structures as harvested rain water is directly put into GW reservoir
15	Putting waste near HP/TW will not cause any problem	Such actions will pollute wells and water



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